Intrinsic invariants of $(3,4)$-cuspidal edges 12月のRIMSに間に合うように，11月中に投稿する！

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Abstract

$(3,4)$-カスプ辺は楕円面などの平行曲面上に自然に現れる特異点である．カスプ辺では特異曲率の符号が局所的な形状と関係するが，$(3,4)$-カスプ辺ではその符号が一意に定まらない．本論文では，$(3,4)$-カスプ辺と $(3,5)$-カスプ辺を含む 3-type edge に対して，第一基本形式から二つの内在的不変量 β, δ を構成する．これらを用いて，ガウス曲率の有界性，$(3,4)$-カスプ辺の凸・鞍状，およびジェネリックな $(3,4)$-カスプ辺と $(3,5)$-カスプ辺の内在的判定条件を与える．さらに，適切な仮定のもとで，これらの特異点型が局所等長性によって保存されることを示す．

1 Introduction

One of the fundamental questions in the study of surfaces with singularities is to what extent the singularity type can be determined by the first fundamental form. One way to approach this question is to ask whether two locally isometric generic frontals must have $\mathcal{A}$-equivalent map-germs at corresponding singular points. We refer to this question as *property-$\mathcal{A}$*. More precisely, we ask whether

$$\boxed{f, \bar{f} \text{ are locally isometric generic frontals } \Longrightarrow\ f \sim_{\mathcal{A}} \bar{f} \quad ?} \tag{1.1}$$

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Several results are known in this direction. Honda–Naokawa–Umehara–Yamada [5, Proposition 4] established intrinsic criteria for cuspidal edges, swallowtails, and cuspidal cross caps at non-degenerate singular points with non-vanishing limiting normal curvature. Consequently, these singularity types are preserved under local isometries, subject to the appropriate assumptions. Honda–Saji [6, Corollary 4.5] further showed that $(2,5)$-cuspidal edges are intrinsically characterized among frontals having singular points of the first kind with non-vanishing limiting normal curvature. In contrast, Hattori–Honda–Morimoto [7, Corollary 4.9] constructed a counterexample to property-$\mathcal{A}$ for isometric generic frontals. In their example, a $(4,5)$-cuspidal edge and a $(4,5,+7)$-cuspidal edge can be locally isometric. Table 1 summarizes these results together with the singularity types considered in the present paper.

Table 1: Preservation of singularity types under local isometries of generic frontals.

Singularity type	Preservation	Reference
Cuspidal edges, swallowtails, and cuspidal cross caps	Yes	Honda–Naokawa–Umehara–Yamada (2020) [5]
$(2,5)$-cuspidal edges	Yes	Honda–Saji (2019) [6]
$(4,5)$-cuspidal edges and $(4,5,\pm 7)$-cuspidal edges	Not in general	Hattori–Honda–Morimoto (2025) [7]
$(3,4)$-cuspidal edges and $(3,5)$-cuspidal edges	Yes	The present paper (Corollary D)

Thus, property-$\mathcal{A}$ does not hold in general; its validity depends on the class of singularities under consideration. To investigate this problem, it is essential to find effective criteria for distinguishing singularity types. In particular, if such criteria can be expressed in terms of invariants determined by the first fundamental form, then their preservation under local isometries can be used to establish the preservation of singularity types. In this paper, we obtain such criteria for $(3,4)$- and $(3,5)$-cuspidal edges and use them to prove that these singularity types satisfy property-$\mathcal{A}$ under suitable assumptions.

We first clarify the assumptions underlying our criteria. A *singular point of degeneracy order* 2, as considered in this paper, is a singular point at which the signed area density function vanishes to order two in a direction transverse to the singular set. On the other hand, a singular point is said to be *of the first kind* if the tangent direction to the singular set is distinct from the null direction. These are independent conditions: degeneracy order 2 does not, by itself, imply that a singular point is of the first kind. Indeed, Figure 1 shows an example of a wave front having a singular point of degeneracy order 2 at the origin that is not of the first kind. In this example, the tangent direction to the singular set coincides with the null direction at the origin. We therefore distinguish these two conditions and focus mainly on singular points of the first kind with degeneracy order 2.

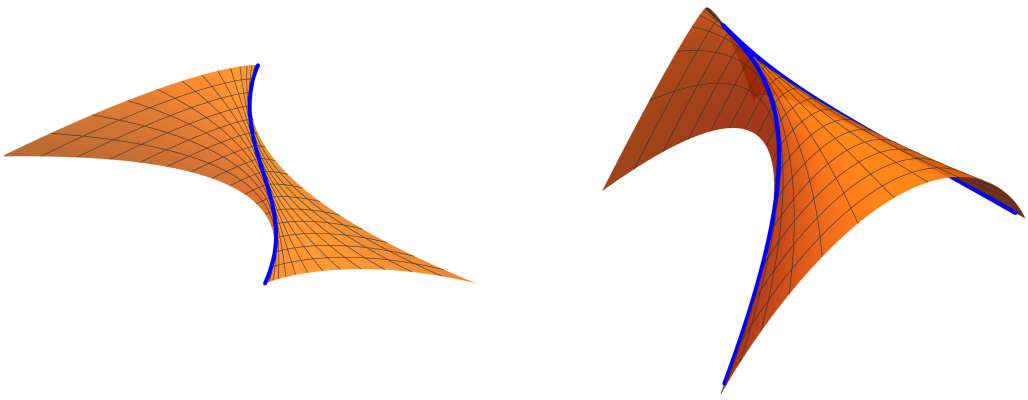


Figure 1: Examples of singular points of degeneracy order 2. Left: a singular point of the first kind. Right: a singular point at the origin that is not of the first kind. The singular sets are shown in blue.

We next explain the geometric motivation for studying $(3,4)$-cuspidal edges. For

example, as the offset distance of the parallel surfaces of an ellipsoid varies, $(3,4)$-cuspidal edges appear at the first stage at which singularities occur, before ordinary cuspidal edges appear (Figure 2). Despite their natural occurrence, the differential geometry of $(3,4)$-cuspidal edges has been studied less extensively than that of ordinary cuspidal edges. In particular, the sign of the singular curvature along a $(3,4)$-cuspidal edge depends on the orientation of the null vector field and hence is not well-defined without an additional choice of orientation. This motivates the search for intrinsic invariants that characterize the local shape of $(3,4)$-cuspidal edges, especially whether they are convex or saddle-shaped.

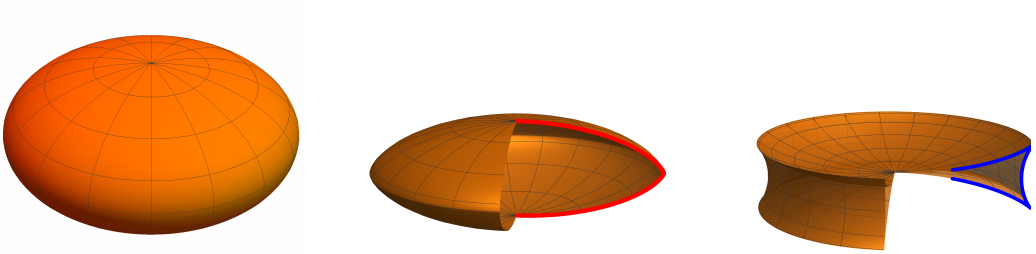


Figure 2: A rotational ellipsoid (left) and two parallel surfaces (center and right). The red line shows a $(3,4)$-cuspidal edge. The blue line shows a cuspidal edge.

For ordinary cuspidal edges, the singular curvature is known to be an intrinsic invariant, and its sign is related to their local shape [4, 9]. Moreover, Martins–Saji–Umehara–Yamada [3] introduced the *product curvature* $\kappa_\Pi := \kappa_\nu \kappa_c$, the product of the limiting normal curvature κ_ν and the cuspidal curvature κ_c. They proved that this quantity is intrinsic and is closely related to the boundedness of Gaussian curvature.

More generally, Martins–Saji–Santos–Teramoto [1] introduced m-type edges as a class of singularities generalizing ordinary cuspidal edges and defined the $(m, m+i)$-cuspidal curvatures $\omega_{m,m+i}$. Since our main objects are $(3,4)$- and $(3,5)$-cuspidal edges, we focus on 3-type edges. For a general 3-type edge, the usual $(3,5)$-cuspidal curvature $\omega_{3,5}$ depends on the choice of null vector field; it becomes well-defined under the lower-order vanishing condition $\omega_{3,4} \equiv 0$. To avoid this vanishing assumption, we introduce a corrected $(3,5)$-cuspidal curvature $\widehat{\omega}_{3,5}$, which is well-defined once the orientation of the null direction has been fixed. We also extend the notion of product curvature from [3] to 3-type edges by defining the $(3,4)$- and $(3,5)$-product curvatures $\Pi_{3,4}$ and $\Pi_{3,5}$ as products involving the signed limiting normal curvature. In contrast, the quantity δ constructed in this paper from the first fundamental form is defined without imposing any lower-order vanishing condition. In particular, when $\omega_{3,4} \equiv 0$, we obtain

$$\Pi_{3,5} = \frac{1}{2}\delta.$$

Thus, unlike the usual $(3,5)$-product curvature, δ can be defined directly from the first fundamental form without imposing the condition $\omega_{3,4} \equiv 0$.

Motivated by these observations, we construct two quantities β and δ from higher-order derivatives of the first fundamental form. These are intrinsic in the sense made precise below: β is independent of the orientation of the null direction, whereas δ changes sign when that orientation is reversed. We then investigate their relationships with the cuspidal curvatures and derive intrinsic criteria for $(3,4)$- and $(3,5)$-cuspidal edges. Under the assumptions that the singular points are of the first kind, have degeneracy order 2, and are generic, these criteria imply property-$\mathcal{A}$ for the two singularity types. We also use β and δ to study the boundedness of Gaussian curvature and the convex or saddle-shaped geometry of $(3,4)$-cuspidal edges.

We now state our main results. Let Σ be a 2-dimensional manifold, let $f : \Sigma \to \mathbb{R}^3$ be a frontal, and let $p \in \Sigma$ be a singular point of the first kind with degeneracy order 2.

Proposition A (The shape of a $(3,4)$-cuspidal edge)**.** *Suppose that f has a $(3,4)$-cuspidal edge at p. Then the following statements hold:*

(1) *If $\beta(p) > 0$, then f is convex at p.*

(2) *If $\beta(p) < 0$, then f is saddle-shaped at p.*

Theorem B (Boundedness of Gaussian curvature)**.** *Let K denote the Gaussian curvature of f and $S(f)$ its singular set. For the quantities β and δ defined in Section 4, the following conditions are equivalent:*

(1) *There exists a neighborhood U of p such that K is bounded on $U \setminus S(f)$.*

(2) *There exists a neighborhood U of p such that $\beta \equiv \delta \equiv 0$ on $S(f) \cap U$.*

Theorem C (Intrinsic criteria for generic singularity types)**.** *Assume that f is generic at p. Then the following conditions are equivalent:*

(1) *f has a $(3,4)$-cuspidal edge at p.*

(2) *$\beta(p) \neq 0$.*

Moreover, the following conditions are equivalent:

(1) *f has a $(3,5)$-cuspidal edge at p.*

(2) *There exists a neighborhood U of p such that $\beta \equiv 0$ on $S(f) \cap U$ and $\delta(p) \neq 0$.*

Corollary D (Preservation of singularity types under isometries)**.** *Let $g : \overline{\Sigma} \to \mathbb{R}^3$ be a frontal, and let $\overline{p} \in \overline{\Sigma}$ be a singular point of the first kind with degeneracy order 2. Assume that f and g are locally isometric near p and $\overline{p}$, respectively, and that they are generic at these points. Then, for $j = 4, 5$, the following conditions are equivalent:*

(1) *f has a $(3,j)$-cuspidal edge at p.*

(2) *g has a $(3,j)$-cuspidal edge at $\overline{p}$.*

The remainder of the paper is organized as follows. In Section 2, we review the necessary background and introduce the corrected $(3,5)$-cuspidal curvature. In Section 3, we define singular points of degeneracy order 2 and normalized adapted coordinate systems (NACS), establish the existence of such coordinate systems, and derive their transformation formulas. In Section 4, we investigate the intrinsic properties of κ_s, β, and δ and the expansion of v^2K, and prove Theorem A and Proposition B. Finally, in Section 5, we relate the cuspidal curvatures to β and δ through the product curvatures, and establish Theorem C and Corollary D.

2 Preliminaries

Throughout this paper, let Σ and $\overline{\Sigma}$ be two-dimensional manifolds, and let $\mathbb{S}^2 \subset \mathbb{R}^3$ denote the unit sphere centered at the origin. We denote by $\langle\ ,\ \rangle$ and $\|\ \|$ the standard inner product and norm on $\mathbb{R}^3$, respectively. Unless otherwise stated, all maps, functions, and vector fields are assumed to be of class C^∞. Partial derivatives with respect to local coordinates (u,v) are denoted by subscripts. For example, for $j \in \mathbb{N}$, we write $f_{v^j} := \partial^j f/\partial v^j$.

2.1 $(3,4)$-cuspidal edges and 3-type edges

In this paper, a singular point is a point at which a map fails to be an immersion. More precisely, given a map $f : \Sigma \to \mathbb{R}^3$, a point $p \in \Sigma$ is called a *singular point* of f if $\operatorname{rank}(df)_p < 2$. The set of all singular points of f is denoted by

$$S(f) := \{q \in \Sigma \mid \operatorname{rank}(df)_q < 2\}.$$

The following equivalence relation plays an important role in the study of singularities.

Definition 2.1. Two map-germs $f : (\Sigma, p) \to (\mathbb{R}^3, f(p))$ and $\overline{f} : (\overline{\Sigma}, \overline{p}) \to (\mathbb{R}^3, \overline{f}(\overline{p}))$ are said to be $\mathcal{A}$-*equivalent* if there exist diffeomorphism-germs $\varphi : (\Sigma, p) \to (\overline{\Sigma}, \overline{p})$ and $\Phi : (\mathbb{R}^3, f(p)) \to (\mathbb{R}^3, \overline{f}(\overline{p}))$ such that

$$\Phi \circ f \circ \varphi^{-1} = \overline{f}.$$

We now recall the definitions of $(3,4)$- and $(3,5)$-cuspidal edges.

Definition 2.2. Let $f : \Sigma \to \mathbb{R}^3$ be a map and let $p \in \Sigma$. We say that f has a $(3,4)$-*cuspidal edge* at p if the map-germ $f : (\Sigma, p) \to (\mathbb{R}^3, f(p))$ is $\mathcal{A}$-equivalent to the map-germ at the origin of

$$f^{\mathrm{std}}_{3,4} : \mathbb{R}^2 \ni (u,v) \longmapsto \left(v^3, v^4, u\right) \in \mathbb{R}^3.$$

Similarly, we say that f has a $(3,5)$-*cuspidal edge* at p if its map-germ at p is $\mathcal{A}$-equivalent to the map-germ at the origin of

$$f^{\mathrm{std}}_{3,5} : \mathbb{R}^2 \ni (u,v) \longmapsto \left(v^3, v^5, u\right) \in \mathbb{R}^3.$$

More generally, we say that f has a 3-*type edge* at p if its map-germ at p is $\mathcal{A}$-equivalent to the map-germ at the origin of

$$\mathbb{R}^2 \ni (u,v) \longmapsto \left(v^3, v^4 a(u,v), u\right) \in \mathbb{R}^3,$$

where $a(u,v)$ is a smooth function.

By definition, both $(3,4)$- and $(3,5)$-cuspidal edges are 3-type edges. The maps $f^{\mathrm{std}}_{3,4}$ and $f^{\mathrm{std}}_{3,5}$ are called the *standard* $(3,4)$-*cuspidal edge* and the *standard* $(3,5)$-*cuspidal edge*, respectively (see Figure 3). The notion of a 3-type edge is the special case $m = 3$ of an m-type edge introduced by Martins–Saji–Santos–Teramoto [1].

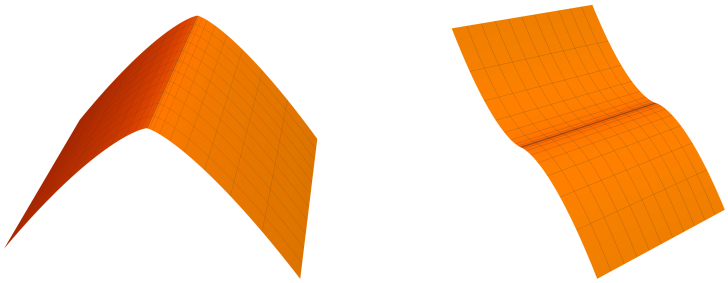


Figure 3: The standard $(3,4)$-cuspidal edge and the standard $(3,5)$-cuspidal edge.

2.2 Criteria for 3-type edges

To state a criterion for 3-type edges, we first introduce some terminology.

Let $f : \Sigma \to \mathbb{R}^3$ be a C^∞ map, and let $p \in S(f)$ be a singular point. Assume that the singular set $S(f)$ is a one-dimensional submanifold (a regular curve) of Σ near p.

For $q \in S(f)$, a nonzero vector $\xi_q \in T_q S(f)$ is called a *singular vector* at q, and the one-dimensional subspace

$$T_q S(f) = \mathrm{span}_{\mathbb{R}}\{\xi_q\} \subset T_q\Sigma$$

is called the *singular direction* at q. A nowhere-vanishing smooth vector field ξ defined on a neighborhood $U \subset \Sigma$ of $S(f)$ is called a *singular vector field* if $\xi_q \in T_q S(f)$ holds for every $q \in S(f) \cap U$.

Suppose further that $q \in S(f)$ is a rank-1 *singular point*, that is, $\mathrm{rank}(df)_q = 1$. A nonzero vector $\eta_q \in \ker(df_q)$ is called a *null vector* at q, and the one-dimensional subspace

$$\ker(df_q) = \mathrm{span}_{\mathbb{R}}\{\eta_q\} \subset T_q\Sigma$$

is called the *null direction* at q. A nowhere-vanishing smooth vector field η defined on a neighborhood $U \subset \Sigma$ of $S(f)$ is called a *null vector field* if $\eta_q \in \ker(df_q)$ holds for every $q \in S(f) \cap U$.

Definition 2.3 (Singular point of the first kind). Let $f : \Sigma \to \mathbb{R}^3$ be a C^∞ map, and assume that the singular set $S(f)$ is a one-dimensional submanifold (a regular curve) of Σ. Let $p \in S(f)$ be a rank-1 singular point. The point p is called a *singular point of the first kind* if the singular direction and the null direction at p are distinct, that is,

$$T_p S(f) \neq \ker(df)_p.$$

A map $f : \Sigma \to \mathbb{R}^3$ is called a *frontal* if there exists a map $\nu : \Sigma \to \mathbb{S}^2$ such that

$$\langle df_q(X_q), \nu(q) \rangle = 0$$

for every $q \in \Sigma$ and every $X_q \in T_q\Sigma$. Such a map ν is called a *unit normal vector field* along f. Furthermore, if the map

$$L := (f, \nu) : \Sigma \longrightarrow \mathbb{R}^3 \times \mathbb{S}^2$$

is an immersion, then f is called a *front* or a *wave front*.

The following criterion for m-type edges, specialized to $m = 3$, is known.

Fact 2.4 ([1, Proposition 2.2])**.** *Let $f : \Sigma \to \mathbb{R}^3$ be a frontal, and let $p \in \Sigma$ be a* rank-1 *singular point. Assume that the singular set $S(f)$ is a regular curve near p, and let ξ be a singular vector field. Suppose further that p is a singular point of the first kind.*

Then f has a 3-type edge at p if and only if there exists a null vector field η satisfying

$$\eta f = \eta^2 f = 0 \quad \text{on } S(f), \tag{2.1}$$

$$\xi f(p) \quad \text{and} \quad \eta^3 f(p) \quad \text{are linearly independent.} \tag{2.2}$$

Here, for a vector field ζ and a map f, the notation $\zeta^i f$ denotes the i-fold directional derivative of f along ζ.

In what follows, we fix a singular vector field ξ and a null vector field η satisfying the conditions of Fact 2.4.

2.3 Cuspidal curvatures and limiting normal curvature

We first recall the notions of invariants, intrinsic invariants, and extrinsic invariants used in this paper.

Definition 2.5. Let $f : \Sigma \to \mathbb{R}^3$ be a frontal, and let $p \in \Sigma$. We denote by $J(f, p)$ a quantity determined by f and p.

(1) The quantity $J(f, p)$ is called an *invariant* if

$$|J(f, p)| = \left|J\left(T \circ f \circ \varphi^{-1}, \varphi(p)\right)\right|$$

holds for every isometry T of $\mathbb{R}^3$ and every diffeomorphism φ of Σ.

(2) Let ds_f^2 and $ds_{\overline{f}}^2$ denote the first fundamental forms of $f : \Sigma \to \mathbb{R}^3$ and $\overline{f} : \overline{\Sigma} \to \mathbb{R}^3$, respectively. We say that f and $\overline{f}$ are *isometric* near p and $\overline{p}$ if there exists a diffeomorphism-germ $\varphi : (\Sigma, p) \to (\overline{\Sigma}, \overline{p})$ such that

$$\varphi^* ds_{\overline{f}}^2 = ds_f^2.$$

(3) An invariant J is called an *intrinsic invariant* if

$$|J(f, p)| = \left|J(\overline{f}, \varphi(p))\right|$$

holds for any two isometric frontals f and $\overline{f}$. An invariant that is not intrinsic is called an *extrinsic invariant*.

For regular surfaces, the Gaussian curvature is an intrinsic invariant, since it is determined by the first fundamental form. On the other hand, the mean curvature is an extrinsic invariant.

(3,4), (3,5)-CEの判定式の直前形に移動する.

We recall the definitions of the $(3, 4)$- and $(3, 5)$-cuspidal curvatures introduced by Martins–Saji–Santos–Teramoto [1].

Definition 2.6. Fix an orientation on a neighborhood U of p. A pair of vector fields (ξ, η) is said to be *positively oriented* if, for every $q \in S(f) \cap U$, the orientation of the ordered basis (ξ_q, η_q) agrees with the chosen orientation of U.

Fact 2.7. *Let $f : \Sigma \to \mathbb{R}^3$ be a frontal having a 3-type edge at $p \in \Sigma$. Suppose that a singular vector field ξ and a null vector field η satisfy the conditions of Fact 2.4. Then,*

$$\omega_{3,4}(p) := \frac{\|\xi f(p)\|^{4/3} \det\big(\xi f(p), \eta^3 f(p), \eta^4 f(p)\big)}{\|\xi f(p) \times \eta^3 f(p)\|^{7/3}} \tag{2.3}$$

is independent of the choice of a positively oriented pair (ξ, η) satisfying the conditions of Fact 2.4 ([1, Proposition 2.10]). The quantity $\omega_{3,4}(p)$ is called the $(3,4)$**-cuspidal curvature** *of f at p.*

Furthermore, suppose that there exists a neighborhood U of p such that $\omega_{3,4}$ vanishes identically on $S(f) \cap U$. Fix an orientation of the null direction along $S(f) \cap U$. Then,

$$\omega_{3,5}(p) := \frac{\|\xi f(p)\|^{5/3} \det\big(\xi f(p), \eta^3 f(p), \eta^5 f(p)\big)}{\|\xi f(p) \times \eta^3 f(p)\|^{8/3}} \tag{2.4}$$

is independent of the choice of a positively oriented pair (ξ, η) satisfying the conditions of Fact 2.4, with η compatible with the chosen orientation ([1, Proposition 2.13]). The quantity $\omega_{3,5}(p)$ is called the $(3,5)$**-cuspidal curvature** *of f at p.*

2.4 Corrected $(3,5)$-cuspidal curvature

The definition of the $(3,5)$-cuspidal curvature $\omega_{3,5}$ is well-defined only when $\omega_{3,4}$ vanishes. We therefore define an invariant that is well-defined even when $\omega_{3,4}$ does not vanish.

Let $f : \Sigma \to \mathbb{R}^3$ be a frontal, and let $p \in \Sigma$ be a 3-type edge. Let $S(f)$ denote the singular set of f. Take a sufficiently small neighborhood U of p and fix an orientation on U. Let (ξ, η) be a positively oriented pair of adapted vector fields such that, along $S(f) \cap U$,

$$\eta f = \eta^2 f = 0, \qquad \operatorname{rank}(\xi f, \eta^3 f) = 2.$$

Along $S(f) \cap U$, set

$$X := \xi f, \qquad Y := \eta^3 f, \qquad Z := \eta^4 f, \qquad W := \eta^5 f.$$

We also define the components of Y and Z orthogonal to X by

$$Y^\perp := Y - \frac{\langle X, Y\rangle}{\|X\|^2} X, \qquad Z^\perp := Z - \frac{\langle X, Z\rangle}{\|X\|^2} X.$$

Since $\operatorname{rank}(X, Y) = 2$, we have $Y^\perp \neq 0$.

Definition 2.8. Define the function

$$\Gamma(\xi, \eta) := \frac{\langle Y^\perp, Z^\perp\rangle}{\|Y^\perp\|^{7/3}} \tag{2.5}$$

along the singular curve. Then

$$\widehat{\omega}_{3,5} := \omega_{3,5} - \frac{5}{3}\Gamma\, \omega_{3,4} \tag{2.6}$$

is called the *corrected* $(3,5)$*-cuspidal curvature.*

Here,

$$\omega_{3,4} = \frac{\|X\|^{4/3} \det(X, Y, Z)}{\|X \times Y\|^{7/3}}, \qquad \omega_{3,5} = \frac{\|X\|^{5/3} \det(X, Y, W)}{\|X \times Y\|^{8/3}}.$$

When $\omega_{3,4} = 0$, we have $\widehat{\omega}_{3,5} = \omega_{3,5}$. Thus, $\widehat{\omega}_{3,5}$ extends the conventional $(3,5)$-cuspidal curvature.

Proposition 2.9. *Fix an orientation of the null direction along the singular curve. Then $\widehat{\omega}_{3,5}$ is independent of the choice of a positively oriented pair of adapted vector fields (ξ, η) satisfying $\eta f = \eta^2 f = 0$ and $\operatorname{rank}(\xi f, \eta^3 f) = 2$.*

Reversing the orientation of the null direction, while preserving the positive orientation of the adapted pair, changes the sign of $\widehat{\omega}_{3,5}$. In particular,

$$|\widehat{\omega}_{3,5}|, \qquad \widehat{\omega}_{3,5} = 0, \qquad \widehat{\omega}_{3,5} \neq 0$$

are independent of the orientation of the null direction.

Proof. We first prove the invariance of $\widehat{\omega}_{3,5}$ under a fixed orientation of the null direction. All calculations below are performed along $S(f) \cap U$. We consider three types of changes of the adapted vector fields: changing ξ, modifying the component of η in the ξ-direction, and multiplying η by a positive smooth function.

Fix the null vector field η. Let $\widetilde{\xi}$ be another singular vector field such that $(\widetilde{\xi}, \eta)$ is positively oriented. Then, along $S(f) \cap U$, we can write $\widetilde{\xi} = c\xi$ for some positive smooth function c. Hence, $\widetilde{X} = cX$, where $\widetilde{X}$ denotes the vector corresponding to $\widetilde{\xi}$. Since $\mathbb{R}X = \mathbb{R}\widetilde{X}$, the orthogonal projection onto the complement of $\mathbb{R}X$ remains unchanged. Thus,

$$\widetilde{Y}^{\perp} = Y^{\perp}, \qquad \widetilde{Z}^{\perp} = Z^{\perp}.$$

It follows from (2.5) that $\widetilde{\Gamma} = \Gamma$. Moreover, since $c > 0$, the definitions give

$$\frac{\|cX\|^{4/3} \det(cX, Y, Z)}{\|cX \times Y\|^{7/3}} = \omega_{3,4}, \qquad \frac{\|cX\|^{5/3} \det(cX, Y, W)}{\|cX \times Y\|^{8/3}} = \omega_{3,5}.$$

Therefore, by (2.6), $\widehat{\omega}_{3,5}$ is independent of the choice of ξ.

Next, we consider a change of the null vector field by adding a component in the ξ-direction. Choose an adapted coordinate system (u, v) such that

$$S(f) \cap U = \{v = 0\}, \qquad \eta = \partial_v.$$

Since the independence of ξ has already been established, we may assume that $\xi = \partial_u$. The conditions $f_v = f_{vv} = 0$ along the singular set imply that

$$f_v = v^2 \psi$$

for some smooth vector-valued function ψ. On the other hand, by the proof of [1, Lemma 2.6], any other null vector field satisfying the same conditions can be written, up to multiplication by a non-vanishing smooth function, in the form

$$E := \partial_v + v^2 a(u, v) \partial_u.$$

Setting $\Phi := \psi + a f_u$, we obtain $Ef = v^2 \Phi$. Since $Ev = 1$, differentiation yields

$$E^3 f = 2\Phi, \qquad E^4 f = 6E\Phi, \qquad E^5 f = 12E^2\Phi$$

along $S(f) \cap U$. Moreover, since $f_{uv} = f_{uvv} = 0$ along the singular set, we have

$$E\Phi = \psi_v + a_v f_u, \qquad E^2\Phi = \psi_{vv} + a_{vv} f_u \quad \text{on } S(f) \cap U.$$

Using $f_{v^3} = 2\psi, f_{v^4} = 6\psi_v, f_{v^5} = 12\psi_{vv}$, we obtain

$$E^3 f = Y + 2aX, \qquad E^4 f = Z + 6a_v X, \qquad E^5 f = W + 12a_{vv} X.$$

Taking the orthogonal projections onto the complement of $\mathbb{R}X$, we find $(E^3 f)^{\perp} = Y^{\perp}, (E^4 f)^{\perp} = Z^{\perp}$. Hence, by (2.5),

$$\Gamma(\xi, E) = \Gamma(\xi, \eta).$$

Furthermore, all additional terms in the expressions for $E^3 f$, $E^4 f$, and $E^5 f$ are multiples of $X = f_u$. Consequently, the quantities $\det(X, Y, Z)$, $\det(X, Y, W)$, and $X \times Y$ remain unchanged. Thus,

$$\omega_{3,4}(\xi, E) = \omega_{3,4}(\xi, \eta), \qquad \omega_{3,5}(\xi, E) = \omega_{3,5}(\xi, \eta).$$

Combining these equalities with (2.6), we conclude that

$$\widehat{\omega}_{3,5}(\xi, E) = \widehat{\omega}_{3,5}(\xi, \eta). \tag{2.7}$$

Therefore, $\widehat{\omega}_{3,5}$ is unchanged by modifying the component of η in the ξ-direction.

Finally, consider $\widetilde{\eta} = h\eta$, where $h > 0$, and set $h_1 := \eta h$. Let $\widetilde{Y}$, $\widetilde{Z}$, and $\widetilde{W}$ denote the vectors corresponding to $\widetilde{\eta}$. Differentiating and using $\eta f = \eta^2 f = 0$ along $S(f) \cap U$, we obtain

$$\widetilde{Y} = h^3 Y, \quad \widetilde{Z} = h^4 Z + 6h^3 h_1 Y, \quad \widetilde{W} = h^5 W + 10h^4 h_1 Z + (25h^3 h_1^2 + 10h^4 \eta^2 h) Y.$$

Substituting these expressions into the definitions of $\omega_{3,4}$ and $\omega_{3,5}$, and using $\|Y^\perp\| = \|X \times Y\|/\|X\|$, we obtain

$$\widetilde{\omega}_{3,4} = \omega_{3,4}, \qquad \widetilde{\omega}_{3,5} = \omega_{3,5} + \frac{10h_1}{h\|Y^\perp\|^{1/3}}\omega_{3,4}.$$

On the other hand, taking the orthogonal projections gives

$$\widetilde{Y}^\perp = h^3 Y^\perp, \qquad \widetilde{Z}^\perp = h^4 Z^\perp + 6h^3 h_1 Y^\perp.$$

Hence, by (2.5),

$$\widetilde{\Gamma} = \Gamma + \frac{6h_1}{h\|Y^\perp\|^{1/3}}.$$

Combining these transformation formulas with (2.6), we obtain

$$\begin{aligned}\widetilde{\widehat{\omega}}_{3,5} - \widehat{\omega}_{3,5} &= \left(10 - \frac{5}{3}\cdot 6\right)\frac{h_1}{h\|Y^\perp\|^{1/3}}\omega_{3,4} \\ &= 0.\end{aligned}$$

Therefore, $\widehat{\omega}_{3,5}$ is invariant under multiplication of η by a positive smooth function.

By the proof of [1, Lemma 2.6], any two null vector fields satisfying the required conditions are locally related by a combination of adding a component in the ξ-direction and positive rescaling. Since $\widehat{\omega}_{3,5}$ is invariant under these changes and is independent of the choice of ξ, it is independent of the choice of the positively oriented adapted pair (ξ, η) when the orientation of the null direction is fixed.

It remains to consider the reversal of the orientation of the null direction. To preserve the positive orientation of the adapted pair, we replace (ξ, η) with $(-\xi, -\eta)$. Then,

$$X \mapsto -X, \qquad Y \mapsto -Y, \qquad Z \mapsto Z, \qquad W \mapsto -W.$$

It follows that $\omega_{3,4}$ remains unchanged, whereas both $\omega_{3,5}$ and Γ change sign. Therefore, by (2.6), $\widehat{\omega}_{3,5}$ also changes sign. □

Remark 2.10. By Proposition 2.9, $\widehat{\omega}_{3,5}$ is a signed invariant when the orientation of the null direction is fixed. Even without fixing this orientation, $|\widehat{\omega}_{3,5}|$ is well-defined.

Moreover, when $\omega_{3,4} = 0$, we have

$$\widehat{\omega}_{3,5} = \omega_{3,5}.$$

Thus, $\widehat{\omega}_{3,5}$ can be regarded as an extension of the $(3,5)$-cuspidal curvature to the case where $\omega_{3,4} \neq 0$.

Remark 2.11. The value of $\widehat{\omega}_{3,5}$ is preserved under orientation-preserving isometries of $\mathbb{R}^3$. Under orientation-reversing isometries, the sign of the determinant is reversed, and hence

$$\widehat{\omega}_{3,5} \longmapsto -\widehat{\omega}_{3,5}.$$

Therefore, $\widehat{\omega}_{3,5}$ is an invariant in the sense adopted in this paper, namely, its absolute value is preserved under coordinate changes in the source and target, as well as under Euclidean isometries.

3 Degeneracy order 2 and normalized adapted coordinate systems

↳定理Cの直後 あたりに移動

In this section, we introduce singular points of degeneracy order 2 and establish the existence of a normalized adapted coordinate system in a neighborhood of such a point, which will be used to compare higher-order transverse derivatives of the first fundamental form.

3.1 Singular points of degeneracy order 2

We define singular points of degeneracy order n in terms of the vanishing order of the signed area density function. Let $f : \Sigma \to \mathbb{R}^3$ be a frontal, $p \in S(f)$ a rank-one singular point, and $\nu : \Sigma \to \mathbb{S}^2$ a unit normal vector field along f. Let $(U; u, v)$ be a local coordinate neighborhood centered at p, and define the signed area density function by

$$\lambda(u,v) := \det\big(f_u(u,v), f_v(u,v), \nu(u,v)\big).$$

Definition 3.1. Let $n \geq 1$ be an integer. A rank-1 singular point $p \in S(f)$ is called a *singular point of degeneracy order n* if there exist smooth real-valued functions $\widehat{\lambda}$ and α defined in a neighborhood of p such that

$$\lambda = \alpha \widehat{\lambda}^n, \qquad d\widehat{\lambda}(p) \neq 0, \qquad \alpha(p) \neq 0.$$

If, in addition, p is a singular point of the first kind in the sense of Definition 2.3, then p is called a *singular point of the first kind with degeneracy order n*.

In particular, singular points of degeneracy order 1 coincide with non-degenerate singular points.

In this paper, we restrict our attention to the case $n = 2$ to study the intrinsic geometry of 3-type edges, particularly $(3,4)$-cuspidal edges and $(3,5)$-cuspidal edges. The relation between singular points of degeneracy order 2 and 3-type edges will be given in Proposition 3.4.

We next show that the degeneracy order is independent of coordinate changes in both the source and target.

Lemma 3.2. *Let $U \subset \Sigma$ be an open set, $f : U \to \mathbb{R}^3$ a frontal, and $\nu : U \to \mathbb{S}^2$ a unit normal vector field along f. Let $\Phi : (\mathbb{R}^3, f(p)) \to (\mathbb{R}^3, \Phi(f(p)))$ be a local diffeomorphism germ, and set $\widetilde{f} := \Phi \circ f$. For each $q \in U$, set $J(q) := d\Phi_{f(q)}$ and define*

$$\widetilde{\nu}(q) := \frac{(J(q)^T)^{-1}\nu(q)}{\|(J(q)^T)^{-1}\nu(q)\|}.$$

Then the signed area density functions λ and $\widetilde{\lambda}$ of f and $\widetilde{f}$, respectively, satisfy

$$\widetilde{\lambda} = \det J \, \|(J^T)^{-1}\nu\| \, \lambda. \tag{3.1}$$

Consequently, local diffeomorphisms of the target preserve the singular set and the degeneracy order.

Proof. Since $f_u \times f_v = \lambda\nu$ and

$$(Ja) \times (Jb) = \det J \, (J^T)^{-1}(a \times b),$$

we obtain

$$\begin{aligned}\widetilde{\lambda} &= \left\langle (Jf_u) \times (Jf_v), \frac{(J^T)^{-1}\nu}{\|(J^T)^{-1}\nu\|} \right\rangle \\ &= \det J \, \|(J^T)^{-1}\nu\| \, \lambda.\end{aligned}$$

Since the coefficient of λ on the right-hand side is a smooth non-vanishing function, the zero set and the vanishing order are preserved. □

Similarly, source coordinate changes preserve the degeneracy order. Thus, we obtain the following corollary.

Corollary 3.3. *Singular points of degeneracy order n are preserved under $\mathcal{A}$-equivalence. In particular, the degeneracy order is independent of coordinate changes in the source and target.*

3.2 Singular points of degeneracy order 2 and 3-type edges

The following relation holds between singular points of degeneracy order 2 and 3-type edges.

Proposition 3.4. *Let $f : \Sigma \to \mathbb{R}^3$ be a frontal, and let $p \in S(f)$ be a singular point. Then*

$$p \text{ is of the first kind with degeneracy order } 2 \iff p \text{ is a 3-type edge of } f.$$

Proof. First, suppose that p is a singular point of the first kind with degeneracy order 2. Since p is of the first kind, we can take a local coordinate system (x, y) near p such that the singular set coincides with the x-axis and ∂_y gives the null direction along the x-axis. Write the first fundamental form in these coordinates as

$$ds_f^2 = \widehat{E}\,dx^2 + 2\widehat{F}\,dx\,dy + \widehat{G}\,dy^2,$$

and set

$$Z_1 := \partial_x, \qquad Z_2 := -\widehat{F}\,\partial_x + \widehat{E}\,\partial_y.$$

Then

$$ds_f^2(Z_1, Z_2) = -\widehat{F}\widehat{E} + \widehat{E}\widehat{F} = 0.$$

Moreover, since $\widehat{E} > 0$ along the x-axis, Z_1 and Z_2 are linearly independent near p. By the local coordinate existence lemma used in the proof of [2, Proposition 2.25], there exists a local coordinate system (u, v) such that

$$\partial_u \parallel Z_1, \qquad \partial_v \parallel Z_2,$$

with $S(f) = \{v = 0\}$. Since $\widehat{F} = \widehat{G} = 0$ along the u-axis, we have $Z_2 = \widehat{E}\partial_y$, so ∂_v gives the null direction. Thus, the first fundamental form in this coordinates can be written as

$$ds_f^2 = E(u, v)\,du^2 + G(u, v)\,dv^2, \qquad G(u, 0) = 0, \qquad E(u, v) > 0. \tag{3.2}$$

By the definition of degeneracy order 2, the signed area density function λ can be written as

$$\lambda = \alpha\widehat{\lambda}^2, \qquad d\widehat{\lambda}(p) \neq 0, \qquad \alpha(p) \neq 0.$$

Since the singular set is the u-axis, we have $\widehat{\lambda}(u, 0) = 0$ and hence $\widehat{\lambda}_u(u, 0) = 0$. Together with $d\widehat{\lambda}(p) \neq 0$, this implies $\widehat{\lambda}_v(p) \neq 0$. Therefore, by the division lemma, there exists a smooth function ρ with $\rho(u, 0) \neq 0$ such that $\widehat{\lambda}(u, v) = v\rho(u, v)$. Since $F = 0$ in (3.2), we have

$$EG = \lambda^2 = v^4\alpha^2\rho^4.$$

Thus, $G = v^4\alpha^2\rho^4/E$. Since $G(u, 0) = 0$, we have $f_v(u, 0) = \mathbf{0}$ and $G_{vv}(u, 0) = 0$. On the other hand, since $G_{vv} = 2\langle f_{vv}, f_{vv}\rangle + 2\langle f_v, f_{v^3}\rangle$, we obtain $f_{vv}(u, 0) = \mathbf{0}$. Hence, by the division lemma, there exists a smooth vector-valued function ψ such that

$$f_v = v^2\psi. \tag{3.3}$$

Furthermore, $\langle \psi, \psi\rangle = \alpha^2\rho^4/E$, so $\psi(p) \neq \mathbf{0}$. Also, $F = 0$ and (3.3) imply $\langle f_u, \psi\rangle = 0$. Since $f_{v^3}(p) = 2\psi(p)$, $f_u(p)$ and $f_{v^3}(p)$ are linearly independent. Thus, taking $\xi = \partial_u$ and $\eta = \partial_v$, we obtain

$$\eta f = \eta^2 f = \mathbf{0} \quad \text{on } S(f), \qquad \operatorname{rank}\big(\xi f(p), \eta^3 f(p)\big) = 2.$$

Therefore, applying [1, Proposition 2.2] with $m = 3$, we conclude that p is a 3-type edge.

Conversely, suppose that p is a 3-type edge. Then we can take a local coordinate system (u, v) near p such that

$$S(f) = \{v = 0\}, \qquad \xi = \partial_u, \qquad \eta = \partial_v,$$

and

$$f_v(u, 0) = f_{vv}(u, 0) = \mathbf{0}, \qquad \operatorname{rank}\big(f_u(p), f_{v^3}(p)\big) = 2.$$

By the division lemma, there exists a smooth vector-valued function ψ such that $f_v = v^2\psi$. Since $f_{v^3}(u,0) = 2\psi(u,0)$, the linear independence of $f_u(p)$ and $f_{v^3}(p)$ implies that $f_u \times \psi \neq \mathbf{0}$ near p. Define

$$\nu := \frac{f_u \times \psi}{\|f_u \times \psi\|}.$$

Then ν is a unit normal vector field along f, and the corresponding signed area density function is given by $\lambda = \det(f_u, f_v, \nu) = v^2\|f_u \times \psi\|$. Setting $\widehat{\lambda} := v$ and $\alpha := \|f_u \times \psi\|$, we obtain

$$\lambda = \alpha\widehat{\lambda}^2, \qquad d\widehat{\lambda}(p) = dv \neq 0, \qquad \alpha(p) \neq 0.$$

Hence, p has degeneracy order 2. Since $f_u(p) \neq \mathbf{0}$, $f_v(p) = \mathbf{0}$ and $S(f) = \{v = 0\}$, the null direction ∂_v is transverse to the singular curve, so p is of the first kind. □

3.3 Introduction of adapted coordinate systems

We introduce an adapted coordinate system for our main results, and discuss its existence and metric properties.

Definition 3.5. Let $f : \Sigma \to \mathbb{R}^3$ be a frontal, $\nu : \Sigma \to \mathbb{S}^2$ a unit normal vector field along f, and $p \in \Sigma$ a singular point of the first kind with degeneracy order 2. A local coordinate system $(U; s, t)$ centered at p is called an *adapted coordinate system* (a.c.s. for short) if the following conditions hold in a sufficiently small neighborhood of p.

(1) $S(f) \cap U = \{(s,t) \in U \mid t = 0\}$.

(2) ∂_t gives the null direction along $S(f)$, and $f_t(s,0) = f_{tt}(s,0) = \mathbf{0}$.

(3) At each $(s,0) \in S(f) \cap U$,

$$\{f_s(s,0), f_{t^3}(s,0), \nu(s,0)\}$$

forms an orthonormal basis of $\mathbb{R}^3$.

Lemma 3.6. *Let $f : \Sigma \to \mathbb{R}^3$ be a frontal, and let $p \in \Sigma$ be a singular point of the first kind with degeneracy order 2. Then there exists an a.c.s. in a neighborhood of p.*

Proof. By Proposition 3.4, we can take a local coordinate system (u, v) near p such that

$$S(f) = \{v = 0\}, \qquad \eta = \frac{\partial}{\partial v},$$

and

$$f_v(u,0) = f_{vv}(u,0) = \mathbf{0}, \qquad f_u(p),\ f_{v^3}(p) \text{ are linearly independent.} \tag{3.4}$$

Since the singular curve $u \mapsto f(u,0)$ is regular, we may take u to be its arc-length parameter so that

$$\|f_u(u,0)\| = 1. \tag{3.5}$$

Moreover, by (3.4) and the division lemma, there exists a smooth vector-valued function φ such that

$$f_v(u,v) = v^2\varphi(u,v). \tag{3.6}$$

In particular, $f_{v^3}(u,0) = 2\varphi(u,0)$. By (3.4) and continuity, we may assume that $f_u(u,0)$ and $f_{v^3}(u,0)$ are linearly independent at every point of the singular curve.

Let $a(u)$ be a smooth function and $r(u)$ a smooth positive function, and set

$$s = u + a(u)v^3, \qquad t = r(u)v. \tag{3.7}$$

Along $v = 0$, the Jacobian matrix of the coordinate change $(u,v) \mapsto (s,t)$ is

$$\frac{\partial(s,t)}{\partial(u,v)}(u,0) = \begin{pmatrix} 1 & 0 \\ 0 & r(u) \end{pmatrix}.$$

Since $r(u) > 0$, (3.7) defines a local diffeomorphism. Moreover, $t = 0$ if and only if $v = 0$, so the singular set is still given by $\{t = 0\}$. By the chain rule and (3.6), there exists a smooth vector-valued function $\widetilde{\varphi}$ such that

$$f_t(s,t) = t^2\widetilde{\varphi}(s,t).$$

Hence,

$$f_t(s,0) = f_{tt}(s,0) = \mathbf{0}, \tag{3.8}$$

and ∂_t gives the null direction along the singular set. Furthermore, applying the chain rule along $t = 0$, we obtain

$$f_s(s,0) = f_u(u,0), \tag{3.9}$$

$$f_{t^3}(s,0) = \frac{1}{r(u)^3}\left\{f_{v^3}(u,0) - 6a(u)f_u(u,0)\right\}. \tag{3.10}$$

Now define

$$a(u) := \frac{1}{6}\left\langle f_u(u,0), f_{v^3}(u,0)\right\rangle, \tag{3.11}$$

$$r(u) := \left\|f_{v^3}(u,0) - \left\langle f_u(u,0), f_{v^3}(u,0)\right\rangle f_u(u,0)\right\|^{1/3}. \tag{3.12}$$

Since $f_u(u,0)$ and $f_{v^3}(u,0)$ are linearly independent and (3.5) holds, the vector inside the norm in (3.12) is nonzero. Thus, r is a smooth positive function. By (3.9)–(3.12), we obtain

$$\|f_s(s,0)\| = 1, \qquad \langle f_s(s,0), f_{t^3}(s,0)\rangle = 0, \qquad \|f_{t^3}(s,0)\| = 1.$$

Moreover, by (3.6), $f_{v^3}(u,0)$ is orthogonal to $\nu(u,0)$, and so is $f_u(u,0)$. Hence, $f_{t^3}(s,0)$ is also orthogonal to $\nu(s,0)$. Therefore,

$$\left\{f_s(s,0), f_{t^3}(s,0), \nu(s,0)\right\}$$

forms an orthonormal basis of $\mathbb{R}^3$. Together with (3.8), this shows that (s,t) is an a.c.s. □

Lemma 3.7. *Let (s,t) be an a.c.s. obtained in Lemma 3.6, and write the first fundamental form as*

$$ds_f^2 = E_0(s,t)\,ds^2 + 2F_0(s,t)\,ds\,dt + G_0(s,t)\,dt^2,$$

where

$$E_0 = \langle f_s, f_s\rangle, \qquad F_0 = \langle f_s, f_t\rangle, \qquad G_0 = \langle f_t, f_t\rangle.$$

Then the following hold along the singular set:

$$\begin{aligned} E_0(s,0) &= 1,\\ F_0(s,0) &= (F_0)_t(s,0) = (F_0)_{tt}(s,0) = 0,\\ G_0(s,0) &= (G_0)_t(s,0) = (G_0)_{tt}(s,0) = (G_0)_{t^3}(s,0) = 0,\\ (G_0)_{t^4}(s,0) &= 6. \end{aligned}$$

Proof. Since (s,t) is an a.c.s., we have

$$f_t(s,0) = f_{tt}(s,0) = \mathbf{0}, \qquad \left\{f_s(s,0), f_{t^3}(s,0), \nu(s,0)\right\} \text{ forms an orthonormal basis of } \mathbb{R}^3. \tag{3.13}$$

In particular,

$$E_0(s,0) = \langle f_s, f_s\rangle(s,0) = 1.$$

Next, consider $F_0 = \langle f_s, f_t\rangle$. By the Leibniz rule, for any integer $j \geq 0$, we have

$$(F_0)_{t^j} = \sum_{k=0}^{j}\binom{j}{k}\left\langle f_{st^k}, f_{t^{j-k+1}}\right\rangle. \tag{3.14}$$

Using (3.13) and (3.14), we obtain along $t = 0$

$$\begin{aligned} F_0(s,0) &= \langle f_s, f_t\rangle(s,0) = 0,\\ (F_0)_t(s,0) &= \langle f_{st}, f_t\rangle(s,0) + \langle f_s, f_{tt}\rangle(s,0) = 0,\\ (F_0)_{tt}(s,0) &= \langle f_{stt}, f_t\rangle(s,0) + 2\langle f_{st}, f_{tt}\rangle(s,0) + \langle f_s, f_{t^3}\rangle(s,0) = 0. \end{aligned}$$

Here, the last equality follows from the orthogonality of $f_s(s,0)$ and $f_{t^3}(s,0)$.

Next, consider $G_0 = \langle f_t, f_t \rangle$. Similarly, the Leibniz rule gives

$$(G_0)_{t^j} = \sum_{k=0}^{j} \binom{j}{k} \langle f_{t^{k+1}}, f_{t^{j-k+1}} \rangle \tag{3.15}$$

for any integer $j \geq 0$. Hence, Using (3.13) and (3.15) along $t = 0$, we have

$$\begin{aligned}
G_0(s,0) &= \langle f_t, f_t \rangle(s,0) = 0, \\
(G_0)_t(s,0) &= 2\langle f_{tt}, f_t \rangle(s,0) = 0, \\
(G_0)_{tt}(s,0) &= 2\langle f_{t^3}, f_t \rangle(s,0) + 2\langle f_{tt}, f_{tt} \rangle(s,0) = 0, \\
(G_0)_{t^3}(s,0) &= 2\langle f_{t^4}, f_t \rangle(s,0) + 6\langle f_{t^3}, f_{tt} \rangle(s,0) = 0, \\
(G_0)_{t^4}(s,0) &= 2\langle f_{t^5}, f_t \rangle(s,0) + 8\langle f_{t^4}, f_{tt} \rangle(s,0) + 6\langle f_{t^3}, f_{t^3} \rangle(s,0) \\
&= 6,
\end{aligned}$$

where the last equality follows from $\|f_{t^3}(s,0)\| = 1$. □

3.4 Construction of normalized adapted coordinate systems

NACS : (x, y) に する。

We establish the existence of a normalized adapted coordinate system by orthogonalizing an a.c.s. using the method of characteristics.

Definition 3.8. Let $f : \Sigma \to \mathbb{R}^3$ be a frontal, and let $p \in \Sigma$ be a singular point of the first kind with degeneracy order 2. Let $(U; u, v)$ be a local coordinate system centered at p such that

$$S(f) \cap U = \{v = 0\},$$

and write the first fundamental form as

$$ds_f^2 = E(u,v)\,du^2 + 2F(u,v)\,du\,dv + G(u,v)\,dv^2,$$

where $E = \langle f_u, f_u \rangle$, $F = \langle f_u, f_v \rangle$, and $G = \langle f_v, f_v \rangle$. The coordinate system (u, v) is called a *normalized adapted coordinate system* (NACS for short) if the following conditions hold:

$$E(u,0) = 1, \qquad F(u,v) \equiv 0, \tag{3.16}$$

$$G(u,0) = G_v(u,0) = G_{vv}(u,0) = G_{v^3}(u,0) = 0, \tag{3.17}$$

$$G_{v^4}(u,0) = 6. \tag{3.18}$$

It follows from the definition of a NACS that $f_v(u,0) = f_{vv}(u,0) = \mathbf{0}$, $\|f_u(u,0)\| = \|f_{v^3}(u,0)\| = 1$, and $\langle f_u, f_{v^3} \rangle(u,0) = 0$. (つまり、3-type edge であると言っている。)

Proposition 3.9. *Let $f : \Sigma \to \mathbb{R}^3$ be a frontal, and let $p \in \Sigma$ be a singular point of the first kind with degeneracy order 2. Then there exists a NACS in a neighborhood of p.*

☆「NACS は a.c.s である」は、別途示すこと

Proof. By Lemma 3.6, we can take an a.c.s. (s, t) and write its first fundamental form using the notation of Lemma 3.7. By the division lemma, there exist smooth real-valued functions A and B such that

$$F_0 = t^3 A, \qquad G_0 = t^4 B, \qquad B(s,0) = \frac{1}{4}. \tag{3.19}$$

Consider new coordinates $(u(s,t), v(s,t))$ with $v = t$. We define u as a local solution of the first-order partial differential equation

$$F_0 u_s - E_0 u_t = 0, \qquad u(s,0) = s. \tag{3.20}$$

This is the initial value problem for

$$u_t - \frac{F_0}{E_0} u_s = 0.$$

Since $F_0 = 0$ and $E_0 = 1$ along $t = 0$, the characteristic direction is transverse to the initial curve $t = 0$. Thus, the method of characteristics guarantees the existence of a local solution. The initial condition gives $u_s(s,0) = 1$, so the coordinate change $(s,t) \mapsto (u,v)$

is a local diffeomorphism after shrinking the neighborhood if necessary. We show that (u, v) is a NACS.

Writing the inverse transformation as $s = s(u, v)$ and $t = v$, we have

$$\partial_u = \frac{1}{u_s}\partial_s, \qquad \partial_v = -\frac{u_t}{u_s}\partial_s + \partial_t.$$

Let E, F, G denote the coefficients of the first fundamental form in the new coordinates. By (3.20), we obtain

$$F = \langle \partial_u, \partial_v \rangle = \frac{-u_t E_0 + u_s F_0}{u_s^2} = 0.$$

Moreover, since $u_s(s, 0) = E_0(s, 0) = 1$, we have $E(u, 0) = 1$. Furthermore,

$$G = \langle \partial_v, \partial_v \rangle = G_0 - \frac{F_0^2}{E_0} = t^4 B - t^6 \frac{A^2}{E_0}. \tag{3.21}$$

Since $v = t$ and $B(s, 0) = 1/4$, conditions (3.17) and (3.18) follow. □

The Taylor expansions of the first fundamental form can be obtained in a NACS.

Corollary 3.10. *Let $f : \Sigma \to \mathbb{R}^3$ be a frontal, let $p \in \Sigma$ be a singular point of the first kind with degeneracy order 2, and let (u, v) be a NACS near p. Then*

$$E_v(u, 0) = E_{vv}(u, 0) = 0 \tag{3.22}$$

holds. Moreover,

$$E(u, v) = 1 + \frac{E_{v^3}(u, 0)}{6} v^3 + \frac{E_{v^4}(u, 0)}{24} v^4 + \frac{E_{v^5}(u, 0)}{120} v^5 + O(v^6), \tag{3.23}$$

$$G(u, v) = \frac{1}{4} v^4 + \frac{G_{v^5}(u, 0)}{120} v^5 + \frac{G_{v^6}(u, 0)}{720} v^6 + O(v^7). \tag{3.24}$$

Proof. Since (u, v) is a NACS, we have $f_v(u, 0) = f_{vv}(u, 0) = \mathbf{0}$. Thus, by the division lemma, we can write $f_v = v^2 \varphi$ for some smooth vector-valued function φ. Hence,

$$E_v = 2\langle f_u, f_{uv} \rangle = O(v^2), \qquad E_{vv} = O(v),$$

which gives $E_v(u, 0) = E_{vv}(u, 0) = 0$. The remaining assertions follow from the NACS conditions (3.16)–(3.18) and Taylor's theorem. □

3.5 Transformation formulas between NACSs

We describe how higher-order derivatives of the coefficients of the first fundamental form transform between two NACSs. Let (u, v) and (ξ, η) be two NACSs near p, and write the coordinate change as

$$u = u(\xi, \eta), \qquad v = v(\xi, \eta).$$

Let E, G and $\widehat{E}, \widehat{G}$ denote the coefficients of the first fundamental form in the coordinates (u, v) and (ξ, η), respectively. Along the singular set, we have

$$v(\xi, 0) = v_\xi(\xi, 0) = 0, \qquad u_\eta(\xi, 0) = 0.$$

Comparing the normalization conditions gives

$$u_\xi(\xi, 0) = \pm 1, \qquad v_\eta(\xi, 0) = \pm 1.$$

Lemma 3.11. *Under the above assumptions, the following hold along the singular set:*

$$\widehat{E}_{\eta^3} = v_\eta E_{v^3}, \tag{3.25}$$

$$\widehat{E}_{\eta^4} = E_{v^4} + 6 v_{\eta^2} E_{v^3}, \tag{3.26}$$

$$\widehat{E}_{\eta^5} = v_\eta E_{v^5} + 10 v_{\eta^2} v_\eta E_{v^4} + (10 v_{\eta^3} + 15\{v_{\eta^2}\}^2 v_\eta) E_{v^3}, \tag{3.27}$$

$$\widehat{G}_{\eta^5} = v_\eta \big(G_{v^5} + 120 v_{\eta^2}\big), \tag{3.28}$$

$$\widehat{G}_{\eta^6} = G_{v^6} + 27 v_{\eta^2} G_{v^5} + 300 v_\eta v_{\eta^3} + 1170\{v_{\eta^2}\}^2. \tag{3.29}$$

Here, the derivatives of $\widehat{E}$, $\widehat{G}$, and v_η are evaluated at $(\xi, 0)$, while those of E and G are evaluated at $(u(\xi, 0), 0)$.

Proof. Comparing the coefficients of the first fundamental forms, we obtain

$$\widehat{E} = Eu_\xi^2 + Gv_\xi^2, \qquad 0 = Eu_\xi u_\eta + Gv_\xi v_\eta, \qquad \widehat{G} = Eu_\eta^2 + Gv_\eta^2.$$

Differentiating the above identities with respect to η to the required orders using the Leibniz rule and evaluating along the singular set, we obtain

$$\begin{aligned}
\widehat{E}_{\eta^3} &= v_\eta E_{v^3} \\
\widehat{E}_{\eta^4} &= E_{\eta^4} + G_{\eta^4}\,(v_\xi)^2 \\
\widehat{E}_{\eta^5} &= E_{\eta^5} + 2u_\xi u_{\xi\eta^5} + G_{\eta^5}\,\big(v_\xi\big)^2 + 10G_{\eta^4}\,v_\xi v_{\xi\eta} \\
\widehat{G}_{\eta^5} &= \big(v_\eta\big)^2 G_{\eta^5} + 10 v_\eta v_{\eta^2}\,G_{\eta^4} \\
\widehat{G}_{\eta^6} &= G_{\eta^6} + 12 v_\eta v_{\eta^2} G_{\eta^5} + 30\big((v_{\eta^2})^2 + v_\eta v_{\eta^3}\big) G_{\eta^4}.
\end{aligned}$$

Since both coordinate systems are NACSs, we apply (3.16)–(3.18) and (3.22) along the singular set. Computing E_{η^3}, E_{η^4}, E_{η^5}, G_{η^5}, and G_{η^6} by the chain rule, we obtain (3.25)–(3.29). □

4 Intrinsic invariants of singular points of degeneracy order 2

In this section, we use the transformation formulas between NACSs to show that κ_s, β, δ are intrinsic invariants. We also prove Proposition 4.7 and Theorem 4.9.

4.1 Intrinsic expression for singular curvature

We define the singular curvature along 3-type edges and express it in terms of the first fundamental form in a NACS.

Definition 4.1. Let $f : \Sigma \to \mathbb{R}^3$ be a frontal with a unit normal vector field ν, and let $\gamma(t)$ be a parametrization of a singular curve consisting of 3-type edges of the first kind. Set $\widehat{\gamma} := f \circ \gamma$ and $\widehat{\nu} := \nu \circ \gamma$. We define the singular curvature along the 3-type edge by

$$\kappa_s(t) := \frac{\det\big(\widehat{\gamma}'(t), \widehat{\gamma}''(t), \widehat{\nu}(t)\big)}{\|\widehat{\gamma}'(t)\|^3}. \tag{4.1}$$

Since the singular curvature depends on the orientation of ν, we fix its orientation in a NACS (u, v) by

$$\nu(u, 0) = f_u(u, 0) \times f_{v^3}(u, 0),$$

where the right-hand side has unit length by (3.16) and (3.18).

Proposition 4.2. *Let $f : \Sigma \to \mathbb{R}^3$ be a frontal, let $p \in \Sigma$ be a singular point of the first kind with degeneracy order 2, and let (u, v) be a NACS. Then the singular curvature κ_s along the singular curve $v = 0$ is given by*

$$\kappa_s(u) = -\frac{1}{2} E_{v^3}(u, 0). \tag{4.2}$$

Consequently, κ_s is an intrinsic invariant.

Proof. Parametrize the singular curve by $\gamma(u) = (u, 0)$. Then $\widehat{\gamma}'(u) = f_u(u, 0)$ and $\widehat{\gamma}''(u) = f_{uu}(u, 0)$. By the NACS conditions,

$$\|f_u(u, 0)\| = \|f_{v^3}(u, 0)\| = 1, \qquad \langle f_u, f_{v^3}\rangle(u, 0) = 0.$$

Moreover, the orientation of the unit normal vector field is chosen so that $\nu(u, 0) = f_u(u, 0) \times f_{v^3}(u, 0)$. Thus, by the definition of singular curvature,

$$\begin{aligned}
\kappa_s(u) &= \det(f_u, f_{uu}, \nu)(u, 0) = \langle f_u \times f_{uu}, f_u \times f_{v^3}\rangle\,(u, 0) \\
&= \langle f_{uu}, f_{v^3}\rangle(u, 0).
\end{aligned} \tag{4.3}$$

On the other hand, since (u,v) is a NACS, we have $f_v = f_{vv} = \mathbf{0}$ along the singular curve. Differentiating with respect to u gives $f_{uv} = f_{uvv} = \mathbf{0}$. Hence, differentiating $E = \langle f_u, f_u \rangle$ three times with respect to v, we obtain

$$E_{v^3}(u,0) = 2\langle f_u, f_{uv^3} \rangle (u,0). \tag{4.4}$$

Furthermore, since $F = \langle f_u, f_v \rangle \equiv 0$, we have

$$0 = F_{uvv}(u,0) = \langle f_{uu}, f_{v^3} \rangle (u,0) + \langle f_u, f_{uv^3} \rangle (u,0).$$

Thus, by (4.4), $E_{v^3}(u,0) = -2\langle f_{uu}, f_{v^3} \rangle (u,0)$. Substituting this into (4.3), we obtain

$$\kappa_s(u) = -\frac{1}{2} E_{v^3}(u,0).$$

By (3.25), κ_s is preserved up to sign under changes of NACS. κ_s is an intrinsic invariant. □

4.2 Introduction of the intrinsic invariants β and δ

We define β and δ using higher-order derivatives of the coefficients of the first fundamental form in a NACS.

Definition 4.3. Let $f : \Sigma \to \mathbb{R}^3$ be a frontal, let $p \in \Sigma$ be a singular point of the first kind with degeneracy order 2, and let (u,v) be a NACS near p. We define the functions β and δ along the singular curve $v = 0$ by

定義を $\beta(u)$, $\delta(u)$ ではなく、$\beta(p)$, $\delta(p)$ にする. $(p=(0,0))$

$$\beta(u) := -E_{v^4}(u,0) + \frac{G_{v^5}(u,0)}{20} E_{v^3}(u,0), \tag{4.5}$$

$$\delta(u) := -E_{v^5}(u,0) + \frac{G_{v^5}(u,0)}{12} E_{v^4}(u,0) - \left(\frac{7G_{v^5}(u,0)^2}{1200} - \frac{G_{v^6}(u,0)}{30} \right) E_{v^3}(u,0). \tag{4.6}$$

Proposition 4.4. *β defined in Definition 4.3 is an intrinsic invariant independent of the orientation of the null direction. δ is invariant under changes of NACS preserving the orientation of the null direction, and changes sign when the null direction is reversed.*

(u,v) と (ξ,η) が NACS であることを明記する.

Proof. Let $\widehat{\beta}$ and $\widehat{\delta}$ denote the corresponding quantities computed in another NACS. Substituting (3.25), (3.26), and (3.28), we obtain

$$\widehat{\beta} = -E_{v^4} - 6v_{\eta^2} E_{v^3} + \frac{v_\eta (G_{v^5} + 120 v_{\eta^2})}{20} v_\eta E_{v^3} = -E_{v^4} + \frac{G_{v^5}}{20} E_{v^3} = \beta.$$

Similarly, substituting (3.25)–(3.29) into (4.6), all terms involving $v_{\eta^2}(\xi,0)$ and $v_{\eta^3}(\xi,0)$ cancel. Thus, $\widehat{\delta} = \delta$ if $v_\eta = 1$, and $\widehat{\delta} = -\delta$ if $v_\eta = -1$. □

4.3 The sign of β and the shape of $(3,4)$-cuspidal edges (~~ジェネリックを仮定から外して、必要条件のみ~~)

We investigate the relationship between the sign of β and the local shape of $(3,4)$-cuspidal edges. We first express β in terms of derivatives of f.

Proposition 4.5. *Let $f : \Sigma \to \mathbb{R}^3$ be a frontal, and let $p \in \Sigma$ be a singular point of the first kind with degeneracy order 2. Let (u,v) be a local coordinate system near p satisfying*

$$S(f) = \{v = 0\}, \qquad f_v(u,0) = f_{vv}(u,0) = \mathbf{0}.$$

Then, along the singular curve,

$$\beta = 2\frac{\det(f_u, f_{v^3}, f_{uu}) \det(f_u, f_{v^3}, f_{v^4})}{\|f_u\|^{2/3} \|f_u \times f_{v^3}\|^{10/3}}. \tag{4.7}$$

In particular, if (u,v) is an a.c.s., then

$$\beta = 2\det(f_u, f_{v^3}, f_{uu}) \det(f_u, f_{v^3}, f_{v^4}). \tag{4.8}$$

判定条件の後に追加

p が 3-type edge の場合:
$\beta > 0 \iff$ convex ジェネリック (3,4)-CE
$(< 0) \iff$ saddle-shape ジェネリック (3,4)-CE

(K_S の場合, $K_S > 0$ でも non-ジェネリックの場合がある)

Proof. We first establish the formula in a NACS. Along the singular curve, we have

$$f_v = f_{vv} = \mathbf{0}, \qquad f_{uv} = f_{uvv} = \mathbf{0}.$$

Since $F = \langle f_u, f_v \rangle \equiv 0$, we obtain

$$E_{v^3} = -2\langle f_{uu}, f_{v^3} \rangle, \quad E_{v^4} = -2\langle f_{uu}, f_{v^4} \rangle, \quad G_{v^5} = 20\langle f_{v^3}, f_{v^4} \rangle.$$

Thus, by the definition of β,

$$\beta = 2\left\{ \langle f_{uu}, f_{v^4} \rangle - \langle f_{uu}, f_{v^3} \rangle \langle f_{v^3}, f_{v^4} \rangle \right\}. \tag{4.9}$$

By the NACS conditions, we have $\|f_u\| = \|f_{v^3}\| = 1$ and $\langle f_u, f_{v^3} \rangle = 0$. Moreover, $E(u,0) = 1$ and $F_{v^3}(u,0) = 0$ imply $\langle f_u, f_{uu} \rangle = \langle f_u, f_{v^4} \rangle = 0$. Hence,

$$\langle f_{uu}, f_{v^4} \rangle - \langle f_{uu}, f_{v^3} \rangle \langle f_{v^3}, f_{v^4} \rangle = \det(f_u, f_{v^3}, f_{uu}) \det(f_u, f_{v^3}, f_{v^4}).$$

Therefore, in a NACS,

$$\beta = 2 \det(f_u, f_{v^3}, f_{uu}) \det(f_u, f_{v^3}, f_{v^4}). \tag{4.10}$$

Next, let (u, v) be a local coordinate system satisfying the assumptions of the proposition, and let (ξ, η) be a NACS. Write the coordinate change as $u = u(\xi, \eta)$, $v = v(\xi, \eta)$. Along the singular curve, we have

$$v_\xi = 0, \qquad u_\eta = u_{\eta\eta} = 0.$$

By the chain rule and $f_v = f_{vv} = \mathbf{0}$, we obtain

$$\begin{aligned} f_\xi &= u_\xi f_u, \\ f_\xi \times f_{\eta^3} &= u_\xi v_\eta^3 f_u \times f_{v^3}, \\ \det(f_\xi, f_{\eta^3}, f_{\xi\xi}) &= u_\xi^3 v_\eta^3 \det(f_u, f_{v^3}, f_{uu}), \qquad (4.11) \\ \det(f_\xi, f_{\eta^3}, f_{\eta^4}) &= u_\xi v_\eta^7 \det(f_u, f_{v^3}, f_{v^4}). \qquad (4.12) \end{aligned}$$

Since (ξ, η) is a NACS, we have $\|f_\xi\| = \|f_\xi \times f_{\eta^3}\| = 1$. Thus,

$$|u_\xi| = \frac{1}{\|f_u\|}, \qquad |v_\eta|^3 = \frac{\|f_u\|}{\|f_u \times f_{v^3}\|}. \tag{4.13}$$

Applying (4.10) in the coordinates (ξ, η) and substituting (4.11), (4.12), and (4.13), we obtain

$$\begin{aligned} \beta &= 2u_\xi^4 v_\eta^{10} \det(f_u, f_{v^3}, f_{uu}) \det(f_u, f_{v^3}, f_{v^4}) \\ &= 2\frac{\det(f_u, f_{v^3}, f_{uu}) \det(f_u, f_{v^3}, f_{v^4})}{\|f_u\|^{2/3} \|f_u \times f_{v^3}\|^{10/3}}. \end{aligned}$$

This proves (4.7). In particular, if (u, v) is an a.c.s., then $\|f_u\| = \|f_u \times f_{v^3}\| = 1$, and (4.8) follows. □

Next, we define convex and saddle-shaped behavior to describe the local shape of $(3,4)$-cuspidal edges, and prove Proposition 4.7.

Definition 4.6. Let $f : \Sigma \to \mathbb{R}^3$ be a frontal, let $p \in \Sigma$ be a 3-type edge, and let $LT_p \subset \mathbb{R}^3$ denote the limiting tangent plane at p. We say that f is *convex* at p if there exists a neighborhood V of p such that

$$f(V) \cap LT_p = \{f(p)\}$$

and $f(V) \setminus \{f(p)\}$ lies entirely on one side of LT_p. On the other hand, we say that f is *saddle-shaped* at p if, for every neighborhood V of p, the image $f(V)$ meets both sides of LT_p.

Proposition 4.7. ステートメントと証明を修正する
Let $f : \Sigma \to \mathbb{R}^3$ be a frontal, and let $p \in \Sigma$ be a $(3,4)$-cuspidal edge. Then the following statements hold:

(1) *If $\beta(p) > 0$, then f is convex at p.*

(2) *If $\beta(p) < 0$, then f is saddle-shaped at p.*

Proof. Since a $(3,4)$-cuspidal edge is a $(3,4)$-type edge in the sense of [1], we apply [1, Proposition 2.9, (2.5)] with $(m,n) = (3,4)$. After an orientation-preserving coordinate change on the source and an orientation-preserving Euclidean isometry of the target, we may assume that $p = (0,0)$, $f(p) = \mathbf{0}$, and the limiting tangent plane at p is $z = 0$, with

$$f(u,v) = \left(u,\ \frac{u^2 a(u)}{2} + \frac{v^3}{6},\ \frac{u^2 b_0(u)}{2} + \frac{v^4 b_4(u,v)}{24}\right), \qquad b_4(0,0) \neq 0.$$

In this normal form, we have

$$\begin{aligned} f_u(p) &= (1,0,0), & f_{v^3}(p) &= (0,1,0), \\ f_{uu}(p) &= (0,a(0),b_0(0)), & f_{v^4}(p) &= (0,0,b_4(0,0)). \end{aligned}$$

Thus,

$$\begin{aligned} \|f_u(p)\| = 1, \qquad & \|f_u(p) \times f_{v^3}(p)\| = 1, \\ \det(f_u, f_{v^3}, f_{uu})(p) = b_0(0), \qquad & \det(f_u, f_{v^3}, f_{v^4})(p) = b_4(0,0). \end{aligned}$$

Hence, by the representation of β in Proposition 4.5, we obtain

$$\beta(p) = 2b_0(0)b_4(0,0). \tag{4.14}$$

Choose the unit normal vector at p to be $\nu(p) = (0,0,1)$. The height function measuring the signed distance from the limiting tangent plane is given by

$$h(u,v) := \langle f(u,v) - f(p),\ \nu(p)\rangle = \frac{u^2 b_0(u)}{2} + \frac{v^4 b_4(u,v)}{24}.$$

Since $b_4(0,0) \neq 0$, (4.14) implies that $b_0(0) \neq 0$ whenever $\beta(p) \neq 0$. Thus, in either case $\beta(p) > 0$ or $\beta(p) < 0$, we may shrink the neighborhood so that $b_0(u)$ and $b_4(u,v)$ each have a constant sign.

If $\beta(p) > 0$, then $b_0(0)$ and $b_4(0,0)$ have the same sign. Hence, $h(u,v)$ has a constant nonzero sign for all sufficiently small $(u,v) \neq (0,0)$. Therefore, the image of f lies entirely on one side of the limiting tangent plane, except at $f(p)$, and f is convex at p.

On the other hand, if $\beta(p) < 0$, then $b_0(0)$ and $b_4(0,0)$ have opposite signs. For sufficiently small nonzero u and v, the values

$$h(u,0) = \frac{u^2 b_0(u)}{2}, \qquad h(0,v) = \frac{v^4 b_4(0,v)}{24}$$

have opposite signs. Thus, the image of f meets both sides of the limiting tangent plane in every neighborhood of p, and f is saddle-shaped at p.

Consequently, by (4.14), $\beta(p) > 0$ implies that f is convex at p, while $\beta(p) < 0$ implies that f is saddle-shaped at p. □

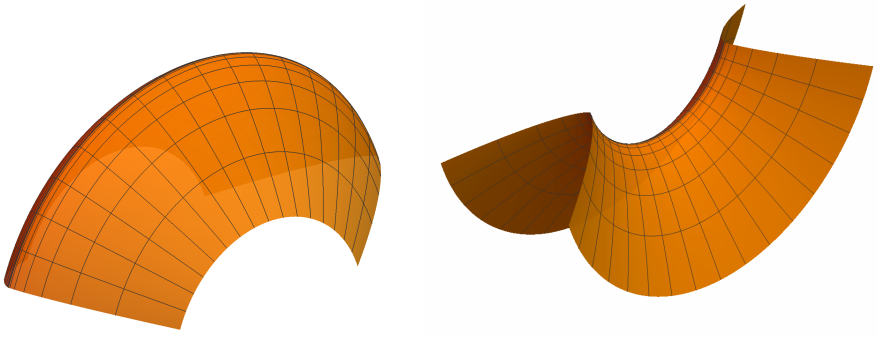


Figure 4: Local shapes of $(3,4)$-cuspidal edges (left: $\beta(p) > 0$; right: $\beta(p) < 0$).

4.4 Boundedness of Gaussian curvature ((3, 4)-カスプ辺の凹凸の後に入れる)

We derive the expansion of v^2K near the singular curve and characterize the boundedness of the Gaussian curvature K in terms of β and δ. Let (u, v) be a NACS, and let K denote the Gaussian curvature of f at regular points. When $F = 0$, Theorema Egregium gives

$$K = \frac{E_v G_v + G_u^2}{4EG^2} + \frac{E_u G_u + E_v^2}{4E^2 G} - \frac{E_{vv} + G_{uu}}{2EG}. \tag{4.15}$$

We set

$$\widetilde{K}(u, v) := v^2 K(u, v).$$

Lemma 4.8. *In a NACS, we have*

$$\widetilde{K}(u, v) = \frac{1}{3}\beta(u) + \left(\frac{1}{6}\delta(u) - \frac{G_{v^5}(u, 0)}{360}\beta(u)\right) v + O(v^2). \tag{4.16}$$

Proof. By (3.23) and (3.24), we have

$$E_v = O(v^2), \quad E_{vv} = O(v), \quad E_u = O(v^3), \qquad G = O(v^4), \quad G_u = G_{uu} = O(v^5).$$

Thus, after multiplying (4.15) by v^2, only the terms

$$\frac{v^2 E_v G_v}{4EG^2}, \qquad -\frac{v^2 E_{vv}}{2EG}$$

contribute to the constant and linear terms, while all remaining terms are $O(v^2)$.

For simplicity, write $E^0_{v^j}(u) := E_{v^j}(u, 0)$ and $G^0_{v^j}(u) := G_{v^j}(u, 0)$ for $j \geq 0$. By Corollary 3.10, the required Taylor expansions are

$$\begin{aligned}
E_v(u, v) &= \frac{1}{2}E^0_{v^3}(u)v^2 + \frac{1}{6}E^0_{v^4}(u)v^3 + \frac{1}{24}E^0_{v^5}(u)v^4 + O(v^5),\\
E_{vv}(u, v) &= E^0_{v^3}(u)v + \frac{1}{2}E^0_{v^4}(u)v^2 + \frac{1}{6}E^0_{v^5}(u)v^3 + O(v^4),\\
G_v(u, v) &= v^3 + \frac{1}{24}G^0_{v^5}(u)v^4 + \frac{1}{120}G^0_{v^6}(u)v^5 + O(v^6),\\
\frac{1}{G(u, v)} &= 4v^{-4}\left\{1 - \frac{G^0_{v^5}(u)}{30}v + \left(\frac{G^0_{v^5}(u)^2}{900} - \frac{G^0_{v^6}(u)}{180}\right)v^2 + O(v^3)\right\},\\
\frac{1}{G(u, v)^2} &= 16v^{-8}\left\{1 - \frac{G^0_{v^5}(u)}{15}v + \left(\frac{G^0_{v^5}(u)^2}{300} - \frac{G^0_{v^6}(u)}{90}\right)v^2 + O(v^3)\right\},\\
\frac{1}{E(u, v)} &= 1 + O(v^3).
\end{aligned}$$

Substituting these expansions, the terms of order v^{-1} cancel, and we obtain

$$\begin{aligned}
\widetilde{K}(u, v) = &-\frac{1}{3}E^0_{v^4}(u) + \frac{G^0_{v^5}(u)}{60}E^0_{v^3}(u)\\
&+\left\{-\frac{1}{6}E^0_{v^5}(u) + \frac{G^0_{v^5}(u)}{60}E^0_{v^4}(u) + \left(\frac{G^0_{v^6}(u)}{180} - \frac{G^0_{v^5}(u)^2}{900}\right)E^0_{v^3}(u)\right\}v + O(v^2).
\end{aligned}$$

Finally, using (4.5) and (4.6), the constant term equals $\beta/3$, while the coefficient of v equals $\delta/6 - G_{v^5}\beta/360$. This proves (4.16). □

The above lemma yields the following theorem 4.9.

Theorem 4.9. *Let $f : \Sigma \to \mathbb{R}^3$ be a frontal, and let $p \in \Sigma$ be a singular point of the first kind with degeneracy order 2. Then the following statements are equivalent:*

(1) *There exists a neighborhood U of p such that K is bounded on $U \setminus S(f)$.*

(2) *There exists a neighborhood U of p such that $\beta \equiv \delta \equiv 0$ on $S(f) \cap U$.*

Proof. By shrinking the neighborhood U of p if necessary, we may assume that a NACS (u, v) is defined on U and that $S(f) \cap U = \{v = 0\}$. By Lemma 4.8, the remainder term can be written using a smooth function $k(u, v)$ as

$$\widetilde{K}(u, v) = \frac{1}{3}\beta(u) + \left(\frac{1}{6}\delta(u) - \frac{G_{v^5}(u, 0)}{360}\beta(u)\right) v + v^2 k(u, v).$$

First, suppose that $\beta \equiv \delta \equiv 0$ on $S(f) \cap U$. Then

$$v^2 K(u, v) = v^2 k(u, v),$$

so $K(u, v) = k(u, v)$ at regular points. By shrinking U further if necessary, we may assume that k is bounded on U. Hence, K is bounded on $U \setminus S(f)$.

Conversely, suppose that K is bounded on $U \setminus S(f)$. For any $(u_0, 0) \in S(f) \cap U$, we have

$$v^2 K(u_0, v) \longrightarrow 0 \qquad (v \longrightarrow 0).$$

Taking $v \to 0$ in the expansion of $\widetilde{K}$, we obtain

$$0 = \frac{1}{3}\beta(u_0).$$

Since u_0 is arbitrary, $\beta \equiv 0$ on $S(f) \cap U$. Dividing the expansion of $\widetilde{K}$ by v, we then obtain

$$vK(u, v) = \frac{1}{6}\delta(u) + vk(u, v).$$

Fixing $u = u_0$ and letting $v \to 0$ again, the boundedness of K implies that the left-hand side tends to zero. Thus, $\delta(u_0) = 0$. Since u_0 is arbitrary, $\delta \equiv 0$ on $S(f) \cap U$. □

Example 4.10. Both of the following maps have $(3, 4)$-cuspidal edges along the singular set $\{v = 0\}$, and the given coordinates satisfy the NACS conditions.

(1)
$$f_0(u, v) = \left(u, \frac{v^3}{6}, \frac{v^4}{24}\right).$$
In this case, $\beta = \delta = 0$, and the Gaussian curvature satisfies $K = 0$ at regular points.

(2)
$$f_{-1}(u, v) = \left(u - \frac{v^7}{1008}, \frac{v^3}{6}\cos u - \frac{v^4}{24}\sin u, \frac{v^3}{6}\sin u + \frac{v^4}{24}\cos u\right).$$
In this case, $\beta = \delta = 0$ as well, and $K \to -1$ as $v \to 0$.

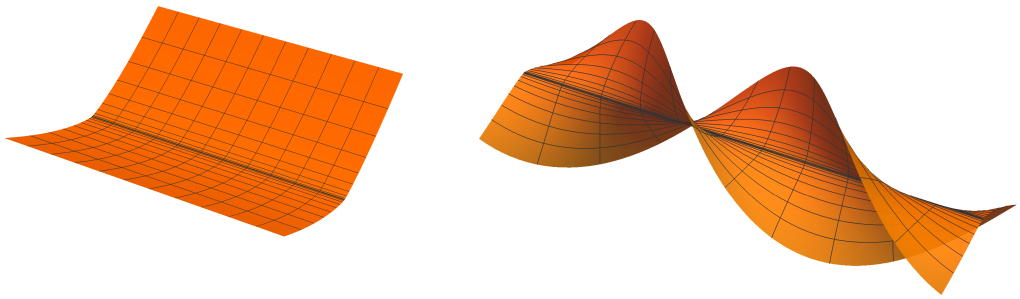


Figure 5: Examples of maps with bounded Gaussian curvature (left: $K = 0$; right: $K \to -1$ near the singular set).

判定条件は 独立したセクションにする。 (5: 判定条件。 6: 積曲率と 定理C。

5 Product curvature and intrinsic criteria

In this section, we introduce the product curvature used in this study and investigate its relationship with β and δ. Combining these results with the criteria for $(3, 4)$- and $(3, 5)$-cuspidal edges, we establish intrinsic criteria for generic $(3, 4)$- and $(3, 5)$-cuspidal edges. As a consequence, we show that these singularity types are preserved under isometries.

5.1 Introduction of signed limiting normal curvature

We introduce the signed limiting normal curvature for 3-type edges.

Definition 5.1. Let $f : \Sigma \to \mathbb{R}^3$ be a frontal with a 3-type edge at $p \in \Sigma$. Let (ξ, η) be a pair of a singular vector field and a null vector field satisfying the conditions of Fact 2.4. We define the signed limiting normal curvature at p by

$$\widetilde{\kappa}_\nu(p) := \frac{\det\big(\xi f(p), \eta^3 f(p), \xi^2 f(p)\big)}{\|\xi f(p) \times \eta^3 f(p)\| \, \|\xi f(p)\|^2}. \tag{5.1}$$

Lemma 5.2. *Let $f : \Sigma \to \mathbb{R}^3$ be a frontal with a 3-type edge. Then $\widetilde{\kappa}_\nu$ is independent of the choice of a positively oriented pair of vector fields (ξ, η) satisfying the conditions of Fact 2.4.*

Proof. Take local coordinates (u, v) adapted to (ξ, η) such that

$$S(f) = \{v = 0\}, \qquad \xi = \partial_u, \qquad \eta = \partial_v$$

along the singular curve. Since f has a 3-type edge, we can write $f_v = v^2\psi$ for some smooth vector-valued function ψ.

Let $(\overline{\xi}, \overline{\eta})$ be another positively oriented pair of vector fields satisfying the same conditions. Along the singular curve, we can write

$$\overline{\xi} = \alpha(u)\, \partial_u$$

for some smooth nonvanishing function α. Moreover, by the proof of [1, Lemma 2.6], we can write

$$\overline{\eta} = h(u, v)\, \big(v^2 a(u, v)\partial_u + \partial_v\big)$$

for some smooth nonvanishing function h and smooth function a. Since $f_v = v^2\psi$, we have $\overline{\eta}^3 f = h^3(f_{v^3} + 2a f_u)$ along the singular curve. On the other hand, we also have $\overline{\xi}^{\,2} f = \alpha^2 f_{uu} + \alpha\alpha' f_u$ along the singular curve. Thus,

$$\det\big(\overline{\xi} f, \overline{\eta}^3 f, \overline{\xi}^{\,2} f\big) = \alpha^3 h^3 \det(f_u, f_{v^3}, f_{uu}),$$

and

$$\|\overline{\xi} f \times \overline{\eta}^3 f\| \, \|\overline{\xi} f\|^2 = |\alpha h^3| \alpha^2 \|f_u \times f_{v^3}\| \, \|f_u\|^2.$$

Therefore,

$$\widetilde{\kappa}_\nu(\overline{\xi}, \overline{\eta}) = \operatorname{sgn}(\alpha h)\, \widetilde{\kappa}_\nu(\xi, \eta).$$

Since both pairs are positively oriented, we have $\alpha h > 0$ along the singular curve. Hence,

$$\widetilde{\kappa}_\nu(\overline{\xi}, \overline{\eta}) = \widetilde{\kappa}_\nu(\xi, \eta),$$

which proves the assertion. □

Remark 5.3. Let $f : \Sigma \to \mathbb{R}^3$ be a frontal, let $p \in \Sigma$ be a rank-one singular point, and let ν be a unit normal vector field along f. Take local coordinates (u, v) such that $f_v(p) = \mathbf{0}$. Since $\operatorname{rank} df_p = 1$, we have $f_u(p) \neq \mathbf{0}$. The limiting normal curvature at p is defined by

$$\kappa_\nu(p) := \frac{\langle f_{uu}(p), \nu(p)\rangle}{\|f_u(p)\|^2}. \tag{5.2}$$

This value is independent of the choice of local coordinates satisfying $f_v(p) = \mathbf{0}$. We call $\kappa_\nu(p)$ the **limiting normal curvature** of f at p [3][4]. It depends on the choice of ν.

For a positively oriented pair of vector fields (ξ, η), choose the unit normal vector field along the singular curve by

$$\nu_{\xi,\eta} := \frac{\xi f \times \eta^3 f}{\|\xi f \times \eta^3 f\|}.$$

Then

$$\widetilde{\kappa}_\nu = \kappa_{\nu_{\xi,\eta}}.$$

Thus, the signed limiting normal curvature can be regarded as the limiting normal curvature whose normal orientation is determined by the pair of singular and null vector fields. In particular, the signed limiting normal curvature is independent of the choice of ν.

Genericity is defined as follows. Let $f : \Sigma \to \mathbb{R}^3$ be a frontal, and let $p \in S(f)$ be a singular point. Suppose that, near p, the singular set $S(f)$ is parametrized by a regular curve $\gamma(t)$ ($|t| < \varepsilon$) satisfying $\gamma(0) = p$. We say that p is **generic** if the following conditions hold:

- $\hat{\gamma}'(0)$ and $\hat{\gamma}''(0)$ are linearly independent, where $\hat{\gamma}(t) := f \circ \gamma(t)$.
- $\mathrm{Span}(\hat{\gamma}'(0), \hat{\gamma}''(0))$ does not coincide with the limiting tangent plane $LT_p := \nu(p)^{\perp}$.

The notion of generic cuspidal edges was introduced in [4], and the above definition extends this notion.

Proposition 5.4. *Let $f : \Sigma \to \mathbb{R}^3$ be a frontal, and let $p \in S(f)$ be a 3-type edge of f. Then p is generic if and only if*

$$\widetilde{\kappa}_\nu(p) \neq 0.$$

Proof. See [10, Proposition 5.2.9]. □

(3.4)-(3.5)-CEの判定条件の後に入れる.

5.2 Introduction of product curvatures

We introduce product curvatures as auxiliary quantities linking the cuspidal curvatures and the signed limiting normal curvature to the intrinsic invariants β and δ.

Definition 5.5. Let $f : \Sigma \to \mathbb{R}^3$ be a frontal, and let $p \in \Sigma$ be a 3-type edge of f. Using $\omega_{3,4}$ and $\widetilde{\kappa}_\nu$ from Definitions 2.7 and 5.1, we define

$$\Pi_{3,4}(p) := \widetilde{\kappa}_\nu(p)\omega_{3,4}(p),$$

which we call the $(3,4)$**-product curvature**. Similarly, using Definition 2.8, we define

$$\Pi_{3,5}(p) := \widetilde{\kappa}_\nu(p)\widehat{\omega}_{3,5}(p),$$

which we call the $(3,5)$**-product curvature**. The condition $\omega_{3,4} \equiv 0$ is not required for this definition.

5.3 Relations between β, δ, and product curvatures

We express the product curvatures in terms of the intrinsic invariants β and δ, thereby establishing their intrinsic nature.

Proposition 5.6. *Let $f : \Sigma \to \mathbb{R}^3$ be a frontal, and let $p \in \Sigma$ be a singular point of the first kind with degeneracy order 2. Then, along the singular curve,*

$$\Pi_{3,4} = \frac{1}{2}\beta.$$

In particular, $\Pi_{3,4}$ is an intrinsic invariant.

Proof. Substituting the definitions of $\widetilde{\kappa}_\nu$ and $\omega_{3,4}$ into Definition 5.5, we obtain

$$\Pi_{3,4} = \frac{\det(f_u, f_{v^3}, f_{uu})\det(f_u, f_{v^3}, f_{v^4})}{\|f_u\|^{2/3}\|f_u \times f_{v^3}\|^{10/3}}.$$

On the other hand, Proposition 4.5 gives

$$\beta = 2\frac{\det(f_u, f_{v^3}, f_{uu})\det(f_u, f_{v^3}, f_{v^4})}{\|f_u\|^{2/3}\|f_u \times f_{v^3}\|^{10/3}}.$$

Hence, $\Pi_{3,4} = \beta/2$, which proves the assertion. □

Proposition 5.7. *Let $f : \Sigma \to \mathbb{R}^3$ be a frontal, and let $p \in \Sigma$ be a singular point of the first kind with degeneracy order 2. Then, along the singular curve,*

$$\Pi_{3,5} = \frac{1}{2}\delta - 2L_v\omega_{3,4}$$

holds. Moreover, if $\beta(p) = 0$ and p is generic, then

$$\Pi_{3,5}(p) = \frac{1}{2}\delta(p).$$

Proof. Take a NACS (u, v) near p, and choose the orientation of the unit normal vector field ν so that $\nu = f_u \times f_{v^3}$ along the singular curve.

By the division lemma, we can write $f_v = v^2\varphi$ for some smooth vector-valued function φ. Define $\widetilde{G}$, $\widetilde{M}$, and $\widetilde{N}$ by

$$\widetilde{G} := \langle \varphi, \varphi \rangle, \qquad \widetilde{M} := \langle \varphi_u, \nu \rangle, \qquad \widetilde{N} := \langle \varphi_v, \nu \rangle.$$

Since $\langle \varphi, \nu \rangle = 0$, we have

$$G = v^4 \widetilde{G}, \qquad M = v^2 \widetilde{M}, \qquad N = v^2 \widetilde{N}.$$

Moreover, since $F \equiv 0$ in a NACS, at regular points we have

$$\widetilde{K} = v^2 K = \frac{L\widetilde{N} - v^2 \widetilde{M}^2}{E\widetilde{G}}. \tag{5.3}$$

Here,

$$L := \langle f_{uu}, \nu \rangle, \quad M := \langle f_{uv}, \nu \rangle, \quad N := \langle f_{vv}, \nu \rangle.$$

In what follows, all quantities are evaluated at $(u, 0)$. Along the singular curve, we have $E = 1$, $E_v = 0$, and $\widetilde{G} = 1/4$. Thus, differentiating (5.3) with respect to v and evaluating at $v = 0$, we obtain

$$\widetilde{K}_v = 4\big(L_v \widetilde{N} + L\widetilde{N}_v\big) - 16L\widetilde{N}\widetilde{G}_v. \tag{5.4}$$

Moreover, since $f_v = v^2\varphi$, we have

$$f_{v^3} = 2\varphi, \qquad f_{v^4} = 6\varphi_v, \qquad f_{v^5} = 12\varphi_{vv}, \qquad \widetilde{\kappa}_\nu = L.$$

By the definition of the corrected $(3, 5)$-cuspidal curvature and the NACS conditions, we obtain

$$\widehat{\omega}_{3,5} = \det(f_u, f_{v^3}, f_{v^5}) - \frac{5}{3}\Gamma\omega_{3,4} = 12\langle \varphi_{vv}, \nu \rangle - \frac{5}{3}\Gamma\omega_{3,4}.$$

Hence,

$$\Pi_{3,5} = 12L\langle \varphi_{vv}, \nu \rangle - \frac{5}{3}\Gamma\Pi_{3,4}. \tag{5.5}$$

On the other hand, since $f_{uv} = 0$ along the singular curve, we have $\langle \nu_v, f_u \rangle = 0$. Also, since ν is a unit normal vector field, $\langle \nu_v, \nu \rangle = 0$. Furthermore, $F \equiv 0$ and $f_v = v^2\varphi$ imply $\langle f_u, \varphi \rangle = \langle \nu, \varphi \rangle = 0$. Thus, ν_v is parallel to φ. Since

$$\langle \nu_v, \varphi \rangle = -\langle \nu, \varphi_v \rangle = -\widetilde{N}, \qquad \|\varphi\|^2 = \widetilde{G} = \frac{1}{4},$$

we obtain $\nu_v = -4\widetilde{N}\varphi$. Therefore,

$$\langle \varphi_v, \nu_v \rangle = -4\widetilde{N}\langle \varphi_v, \varphi \rangle = -2\widetilde{G}_v\widetilde{N},$$

and hence

$$\widetilde{N}_v = \langle \varphi_{vv}, \nu \rangle + \langle \varphi_v, \nu_v \rangle = \langle \varphi_{vv}, \nu \rangle - 2\widetilde{G}_v\widetilde{N}. \tag{5.6}$$

Substituting (5.6) into (5.4) and using

$$\widetilde{K} = 4L\widetilde{N}, \qquad \omega_{3,4} = 6\widetilde{N},$$

together with (5.5), we obtain

$$\Pi_{3,5} = 3\widetilde{K}_v - 2L_v\omega_{3,4} + 18\widetilde{K}\,\widetilde{G}_v - \frac{5}{3}\Gamma\Pi_{3,4}. \tag{5.7}$$

By the NACS conditions and $F \equiv 0$, we have $\Gamma = \langle f_{v^3}, f_{v^4} \rangle = 12\langle \varphi, \varphi_v \rangle = 6\widetilde{G}_v$. Moreover,

$$\Pi_{3,4} = L\omega_{3,4} = 6L\widetilde{N} = \frac{3}{2}\widetilde{K}.$$

Thus, $5\Gamma\Pi_{3,4}/3 = 15\widetilde{K}\,\widetilde{G}_v$. Substituting this into (5.7), we obtain

$$\Pi_{3,5} = 3\widetilde{K}_v - 2L_v\omega_{3,4} + 3\widetilde{K}\,\widetilde{G}_v. \tag{5.8}$$

By Lemma 4.8, we have

$$\widetilde{K}(u,0) = \frac{1}{3}\beta(u), \qquad \widetilde{K}_v(u,0) = \frac{1}{6}\delta(u) - \frac{G_{v^5}(u,0)}{360}\beta(u).$$

Also, differentiating $G = v^4\widetilde{G}$ five times with respect to v and evaluating at $v = 0$, we obtain $G_{v^5} = 120\widetilde{G}_v$. Substituting these identities into (5.8), we obtain

$$\Pi_{3,5} = \frac{1}{2}\delta - \frac{G_{v^5}}{120}\beta - 2L_v\omega_{3,4} + \frac{G_{v^5}}{120}\beta = \frac{1}{2}\delta - 2L_v\omega_{3,4}.$$

Finally, if $\beta(p) = 0$ and p is generic, then $\Pi_{3,5}(p) = \delta(p)/2$, which completes the proof. □

Remark 5.8. The classical $(3,5)$-cuspidal curvature $\omega_{3,5}$ is well-defined under the condition $\omega_{3,4} \equiv 0$. In contrast, δ, introduced in this paper, does not require this condition and is determined up to sign by the first fundamental form. Moreover, if $\beta(p) = 0$ and p is generic, then $\delta(p) = 2\widetilde{\kappa}_\nu(p)\widehat{\omega}_{3,5}(p)$.

5.4 Criteria for $(3,4)$- and $(3,5)$-cuspidal edges

We present two criteria for singularity types that will be used in the subsequent intrinsic characterization. We first recall Whitney's lemma, which is used in the proofs.

Fact 5.9 ([8])**.** *Let $f(u,v)$ be a C^∞ function defined near the origin in $\mathbb{R}^2$. If $f(u,v) = f(u,-v)$, then there exists a C^∞ function $g(u,v)$ defined near the origin such that $f(u,v) = g(u,v^2)$. On the other hand, if $f(u,v) = -f(u,-v)$, then there exists a C^∞ function $h(u,v)$ defined near the origin such that $f(u,v) = vh(u,v^2)$.*

By Whitney's lemma, any smooth function $C(u,v)$ can be written as

$$C(u,v) = C_0(u^2,v) + uC_1(u^2,v) \tag{5.9}$$

for some smooth functions C_0 and C_1. We repeatedly apply this decomposition in the proofs of Theorems 5.11 and 5.12.

We first establish a common preliminary normal form that follows solely from the assumption that the singular point is of the first kind with degeneracy order 2.

Lemma 5.10. *Let $f : \Sigma \to \mathbb{R}^3$ be a frontal, and let $p \in \Sigma$ be a singular point of the first kind with degeneracy order 2. Then, by choosing suitable local coordinates on the source and target so that $p = (0,0)$ and $f(p) = \mathbf{0}$, we can write*

$$f(u,v) = \big(v^3,\, v^4b_1(u) + v^5b_2(u) + v^6b_3(u,v),\, u\big) \tag{5.10}$$

for some smooth functions $b_1(u)$, $b_2(u)$, and $b_3(u,v)$.

Proof. By taking an a.c.s. as in Definition 3.5, we obtain local coordinates (u,v) satisfying

$$S(f) = \{v = 0\}, \qquad f_v(u,0) = f_{vv}(u,0) = \mathbf{0}, \tag{5.11}$$

with ∂_v giving the null direction. Since $f_u(u,0) \neq \mathbf{0}$ in an a.c.s., the image of the singular curve $u \mapsto f(u,0)$ is regular. Thus, by a suitable change of target coordinates, we may assume that

$$f(u,0) = (0,0,u). \tag{5.12}$$

Write $f = (f_1, f_2, f_3)$. By (5.12), we can write $f_3(u,v) = u + vh(u,v)$ for some smooth function h. Moreover, (5.11) implies $h(u,0) = h_v(u,0) = 0$. Thus,

$$(u,v) \longmapsto (f_3(u,v), v)$$

is a local diffeomorphism whose differential along the singular curve agrees with that of the identity map. This coordinate change also preserves $f_v(u,0) = f_{vv}(u,0) = \mathbf{0}$. Denoting the new coordinates again by (u,v), we have

$$f(u,v) = (f_1(u,v), f_2(u,v), u).$$

By the division lemma, there exist smooth functions a and b such that

$$f(u,v) = \big(v^3 a(u,v), v^3 b(u,v), u\big). \tag{5.13}$$

We now show that $a(u,0)$ and $b(u,0)$ cannot both vanish. Let ν be a unit normal vector field and λ the signed area density function. By the definition of a singular point with degeneracy order 2, we can write

$$\lambda = \alpha \widehat{\lambda}^2, \qquad d\widehat{\lambda} \neq 0, \qquad \alpha \neq 0.$$

Since $f_v = f_{vv} = \mathbf{0}$ along $v = 0$, differentiating $f_v(u,0) = \mathbf{0}$ with respect to u gives $f_{uv} = f_{uuv} = \mathbf{0}$. Hence,

$$\lambda_{uu}(u,0) = 0, \qquad \lambda_{vv}(u,0) = \det\big(f_u(u,0), f_{v^3}(u,0), \nu(u,0)\big).$$

On the other hand, differentiating $\lambda = \alpha \widehat{\lambda}^2$ gives, along $v = 0$,

$$\lambda_{uu} = 2\alpha(\widehat{\lambda}_u)^2, \qquad \lambda_{vv} = 2\alpha(\widehat{\lambda}_v)^2.$$

Thus, $\widehat{\lambda}_u = 0$. Since $d\widehat{\lambda} \neq 0$, we have $\widehat{\lambda}_v \neq 0$, and hence $\lambda_{vv} \neq 0$. Therefore, $f_{v^3}(u,0) \neq \mathbf{0}$, and (5.13) shows that at least one of $a(u,0)$ and $b(u,0)$ is nonzero.

Interchanging the first two target coordinates if necessary, we may assume that $a(u,0) \neq 0$. After shrinking the neighborhood, a is nowhere zero, so

$$s := u, \qquad t := v\sqrt[3]{a(u,v)}$$

defines a local coordinate system. Denoting these coordinates again by (u,v), we obtain

$$f(u,v) = \Big(v^3,\ v^3 \frac{b(u,v)}{a(u,v)},\ u\Big).$$

Expanding b/a in a Taylor series with respect to v at $v = 0$, we can write

$$\frac{b(u,v)}{a(u,v)} = b_0(u) + v b_1(u) + v^2 b_2(u) + v^3 b_3(u,v)$$

for some smooth functions b_0, b_1, b_2, b_3. Applying the target coordinate change $(X,Y,Z) \longmapsto (X,\ Y - X b_0(Z),\ Z)$, we obtain

$$f(u,v) = \big(v^3,\ v^4 b_1(u) + v^5 b_2(u) + v^6 b_3(u,v),\ u\big),$$

which completes the proof. □

Theorem 5.11. *Let $f : \Sigma \to \mathbb{R}^3$ be a frontal, and let $p \in \Sigma$. Then the following statements are equivalent:*

(1) p is a $(3,4)$-cuspidal edge of f.

(2) f is a front in a neighborhood of p, and p is a singular point of the first kind with degeneracy order 2.

Proof. We first prove (1) ⇒ (2). Write the standard $(3,4)$-cuspidal edge as $f_0(u,v) := (v^3, v^4, u)$, in accordance with the roles of the coordinates in an a.c.s. Since $(f_0)_u = (0,0,1)$ and $(f_0)_v = (3v^2, 4v^3, 0)$, we have

$$(f_0)_u \times (f_0)_v = v^2(-4v, 3, 0).$$

Thus, $S(f_0) = \{v = 0\}$. Moreover,

$$\nu_0(u,v) := \frac{(-4v, 3, 0)}{\sqrt{16v^2 + 9}}$$

is a unit normal vector field along f_0. For the Legendrian lift $L_0 := (f_0, \nu_0)$, we have

$$(L_0)_u = ((0,0,1), \mathbf{0}), \qquad (L_0)_v = \left(\mathbf{0}, \left(-\frac{4}{3}, 0, 0\right)\right)$$

along $v = 0$. Since these vectors are linearly independent, f_0 is a front. Furthermore, the signed area density function is given by

$$\lambda_0(u,v) = \det((f_0)_u, (f_0)_v, \nu_0) = v^2\sqrt{16v^2+9}.$$

Setting $\widehat{\lambda}_0(u,v) := v$ and $\alpha_0(u,v) := \sqrt{16v^2+9}$, we obtain

$$\lambda_0 = \alpha_0\widehat{\lambda}_0^2, \qquad d\widehat{\lambda}_0 = dv \neq 0, \qquad \alpha_0(u,0) = 3 \neq 0.$$

Thus, every point along $v = 0$ is a singular point with degeneracy order 2. Moreover, since

$$\ker d(f_0)_{(u,0)} = \mathrm{span}_{\mathbb{R}}\{\partial_v\}, \qquad T_{(u,0)}S(f_0) = \mathrm{span}_{\mathbb{R}}\{\partial_u\},$$

these singular points are of the first kind. Since the front property, the first-kind condition, and degeneracy order 2 are preserved under $\mathcal{A}$-equivalence, every $(3,4)$-cuspidal edge satisfies (2).

Next, we prove (2) $\Rightarrow$ (1). We may assume that p is the origin and $f(p) = \mathbf{0}$. By Lemma 5.10, there exist local coordinates (u,v) and smooth functions $b_1(u)$, $b_2(u)$, and $b_3(u,v)$ such that

$$f(u,v) = \big(v^3,\ v^4b_1(u) + v^5b_2(u) + v^6b_3(u,v),\ u\big). \tag{5.14}$$

Since f is a front, we have $b_1(0) \neq 0$. Indeed, normalizing the normal vector field obtained from $(f_v/v^2) \times f_u$, we see that the first component of its v-derivative along $v = 0$ is a nonzero constant multiple of $b_1(u)$. On the other hand, since $f_v(u,0) = \mathbf{0}$ and $f_u(u,0) \neq \mathbf{0}$, the Legendrian lift has rank 2 if and only if $\nu_v(u,0) \neq \mathbf{0}$. Hence, $b_1(0) \neq 0$. After shrinking the neighborhood, we may assume that $b_1(u) \neq 0$. Applying the target coordinate change $(X,Y,Z) \longmapsto (X, Y/b_1(Z), Z)$, we may write

$$f(u,v) = \big(v^3, v^4 + v^5c(u) + v^6d(u,v), u\big).$$

Now apply the source coordinate change

$$v = \widetilde{v} - \frac{c(u)}{4}\widetilde{v}^2 - \frac{c(u)^2}{16}\widetilde{v}^3,$$

followed by the target coordinate change

$$(X,Y,Z) \longmapsto \left(X + \frac{3c(Z)}{4}Y,\ Y,\ Z\right).$$

Then, denoting the new coordinates again by (u,v), we can write

$$f(u,v) = (v^3, v^4, u) + v^6(A(u,v), B(u,v), 0) \tag{5.15}$$

for some smooth functions A and B. We now apply Whitney's lemma twice to A and B. There exist smooth functions A_j and B_j $(j = 0,1,2,3)$ such that

$$\begin{aligned} A(u,v) &= A_0(v^4,u) + vA_1(v^4,u) + v^2A_2(v^4,u) + v^3A_3(v^4,u),\\ B(u,v) &= B_0(v^4,u) + vB_1(v^4,u) + v^2B_2(v^4,u) + v^3B_3(v^4,u). \end{aligned}$$

Setting $X = v^3$, $Y = v^4$, and $Z = u$, we have

$$v^6 = X^2, \qquad v^7 = XY, \qquad v^8 = Y^2, \qquad v^9 = X^3.$$

Define smooth functions P and Q on the target by

$$\begin{aligned} P(X,Y,Z) &:= X^2A_0(Y,Z) + XYA_1(Y,Z) + Y^2A_2(Y,Z) + X^3A_3(Y,Z),\\ Q(X,Y,Z) &:= X^2B_0(Y,Z) + XYB_1(Y,Z) + Y^2B_2(Y,Z) + X^3B_3(Y,Z). \end{aligned}$$

Then

$$P(v^3,v^4,u) = v^6A(u,v), \qquad Q(v^3,v^4,u) = v^6B(u,v).$$

Thus, defining the target coordinate change by

$$\Phi(X,Y,Z) := (X + P(X,Y,Z), Y + Q(X,Y,Z), Z),$$

we obtain

$$\Phi \circ f_0 = (v^3, v^4, u) + v^6(A(u,v), B(u,v), 0) = f,$$

where $f_0(u,v) = (v^3, v^4, u)$. Since $(dP)_{(0,0,0)} = (dQ)_{(0,0,0)} = 0$, we have $d\Phi_{(0,0,0)} = I_3$. By the inverse function theorem, Φ is a local diffeomorphism near the origin. Therefore, the original map-germ is $\mathcal{A}$-equivalent to f_0, and hence p is a $(3,4)$-cuspidal edge. □

Theorem 5.12. *Let $f : \Sigma \to \mathbb{R}^3$ be a frontal, and let $p \in \Sigma$ be a singular point of the first kind with degeneracy order 2. Then the following statements are equivalent:*

(1) p is a $(3,5)$-cuspidal edge of f.

(2) Along the singular curve near p, $\omega_{3,4} \equiv 0$ and $\omega_{3,5}(p) \neq 0$.

Proof. We first prove (1) ⇒ (2). Consider the standard $(3,5)$-cuspidal edge $f_0(u,v) := (v^3, v^5, u)$. Since

$$(f_0)_u \times (f_0)_v = v^2(-5v^2, 3, 0),$$

we have $S(f_0) = \{v = 0\}$. As in Theorem 5.11, the null direction is ∂_v, the singular direction is ∂_u, and the signed area density function is the product of v^2 and a nonvanishing function. Thus, every point along $v = 0$ is a singular point of the first kind with degeneracy order 2. Taking $\xi_0 := \partial_u$ and $\eta_0 := \partial_v$, we have

$$\xi_0 f_0 = (0,0,1), \quad \eta_0^3 f_0 = (6,0,0), \quad \eta_0^4 f_0 = \mathbf{0}, \quad \eta_0^5 f_0 = (0,120,0)$$

along $v = 0$. Hence,

$$\omega_{3,4} \equiv 0, \qquad \omega_{3,5} = \frac{720}{6^{8/3}} \neq 0.$$

Next, we show that (2) also holds for any $(3,5)$-cuspidal edge f. Since $\omega_{3,4}$ is an invariant and $\omega_{3,5}$ is an invariant under the condition $\omega_{3,4} \equiv 0$, their vanishing and nonvanishing are preserved under local diffeomorphisms of the source and isometries of the target. It therefore suffices to show that these conditions are preserved under local diffeomorphisms of the target. This follows from [1, Lemma 2.3] with $m = 3$ and $n = 5$. Thus, every $(3,5)$-cuspidal edge satisfies (2).

Next, we prove (2) ⇒ (1). By Lemma 5.10, choosing suitable local coordinates on the source and target, we can write

$$f(u,v) = \big(v^3,\, v^4 b_1(u) + v^5 b_2(u) + v^6 b_3(u,v),\, u\big) \tag{5.16}$$

for some smooth functions b_1, b_2, b_3. Taking $\xi = \partial_u$ and $\eta = \partial_v$ along the singular curve, we have $f_u(u,0) = (0,0,1)$ and $f_{v^3}(u,0) = (6,0,0)$. Direct computation gives

$$\omega_{3,4}(u,0) = \frac{144}{6^{7/3}}\, b_1(u),$$
$$\omega_{3,5}(u,0) = \frac{720}{6^{8/3}}\, b_2(u) \quad \text{provided that } b_1(u) = 0.$$

Thus, $\omega_{3,4} \equiv 0$ implies $b_1 \equiv 0$, while $\omega_{3,5}(p) \neq 0$ implies $b_2(0) \neq 0$. After shrinking the neighborhood, we may assume that $b_2(u) \neq 0$. Applying the target coordinate change $(X,Y,Z) \longmapsto (X, Y/b_2(Z), Z)$ and expanding b_3/b_2 around $v = 0$, we can write

$$f(u,v) = \big(v^3,\, v^5 + v^6 c_1(u) + v^7 c_2(u) + v^8 c_3(u,v),\, u\big).$$

To eliminate the v^7 term, apply the source coordinate change

$$v = \widetilde{v} - \frac{c_2(u)}{5}\widetilde{v}^3 - \frac{c_2(u)^2}{25}\widetilde{v}^5.$$

Writing $f = (f_1, f_2, f_3)$, Taylor expansion of (5.16) after these coordinate changes gives

$$f_1(u,\widetilde{v}) = \widetilde{v}^3 - \frac{3c_2(u)}{5}\widetilde{v}^5 + \widetilde{v}^8 A(u,\widetilde{v}),$$
$$f_2(u,\widetilde{v}) = \widetilde{v}^5 + c_1(u)\widetilde{v}^6 + \widetilde{v}^8 B(u,\widetilde{v})$$

for some smooth functions A and B. Denoting $\widetilde{v}$ again by v, we apply Whitney's lemma three times to A and B. There exist smooth functions A_j and B_j $(j = 0, \ldots, 7)$ such that

$$A(u,v) = \sum_{j=0}^{7} v^j A_j(v^8, u), \quad B(u,v) = \sum_{j=0}^{7} v^j B_j(v^8, u).$$

Setting $X = v^3$, $Y = v^5$, and $Z = u$, we have

$$v^8 = XY, \qquad v^9 = X^3, \qquad v^{10} = Y^2, \qquad v^{11} = X^2Y,$$
$$v^{12} = X^4, \qquad v^{13} = XY^2, \qquad v^{14} = X^3Y, \qquad v^{15} = Y^3.$$

Define smooth functions P and Q on the target by

$$\begin{aligned} P(X,Y,Z) &:= XYA_0(XY,Z) + X^3A_1(XY,Z) + Y^2A_2(XY,Z) + X^2YA_3(XY,Z) \\ &\quad + X^4A_4(XY,Z) + XY^2A_5(XY,Z) + X^3YA_6(XY,Z) + Y^3A_7(XY,Z), \\ Q(X,Y,Z) &:= XYB_0(XY,Z) + X^3B_1(XY,Z) + Y^2B_2(XY,Z) + X^2YB_3(XY,Z) \\ &\quad + X^4B_4(XY,Z) + XY^2B_5(XY,Z) + X^3YB_6(XY,Z) + Y^3B_7(XY,Z). \end{aligned}$$

Then

$$P(v^3, v^5, u) = v^8A(u,v), \qquad Q(v^3, v^5, u) = v^8B(u,v).$$

Thus, defining the target coordinate change by

$$\Phi(X,Y,Z) := \left(X - \frac{3c_2(Z)}{5}Y + P(X,Y,Z),\ Y + X^2c_1(Z) + Q(X,Y,Z),\ Z\right),$$

we obtain, for the standard form $f_0(u,v) = (v^3, v^5, u)$,

$$\begin{aligned} \Phi \circ f_0(u,v) &= \left(v^3 - \frac{3c_2(u)}{5}v^5 + v^8A(u,v),\ v^5 + c_1(u)v^6 + v^8B(u,v),\ u\right) \\ &= f(u,v). \end{aligned}$$

Moreover, $(dP)_{(0,0,0)} = (dQ)_{(0,0,0)} = 0$. Hence,

$$d\Phi_{(0,0,0)} = \begin{pmatrix} 1 & -3c_2(0)/5 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \det d\Phi_{(0,0,0)} = 1.$$

By the inverse function theorem, Φ is a local diffeomorphism near the origin. Therefore, the original map-germ is $\mathcal{A}$-equivalent to f_0, and hence p is a $(3,5)$-cuspidal edge. □

5.5 Intrinsic criteria and their preservation under isometries

We combine the preceding results to prove Theorem 5.13 and derive Corollary 5.14 as a consequence.

Theorem 5.13. *Let $f : \Sigma \to \mathbb{R}^3$ be a frontal, and let $p \in \Sigma$ be a singular point of the first kind with degeneracy order 2. Assume that f is generic at p. Then the following statements are equivalent:*

(1) *f has a $(3,4)$-cuspidal edge at p.*

(2) *$\beta(p) \neq 0$.*

Moreover, the following statements are equivalent:

(1) *f has a $(3,5)$-cuspidal edge at p.*

(2) *There exists a neighborhood U of p such that $\beta \equiv 0$ on $S(f) \cap U$ and $\delta(p) \neq 0$.*

Proof. By Proposition 5.6, we have

$$\beta(p) \neq 0 \quad \Longleftrightarrow \quad \Pi_{3,4}(p) \neq 0.$$

Since p is generic, $\widetilde{\kappa}_\nu(p) \neq 0$. Thus, $\beta(p) \neq 0$ is equivalent to $\omega_{3,4}(p) \neq 0$. By [1, Proposition 2.11], a 3-type edge is a front if and only if $\omega_{3,4} \neq 0$. Combining this with Theorem 5.11, we obtain the equivalence for $(3,4)$-cuspidal edges.

Next, we consider $(3,5)$-cuspidal edges. Since p is generic, we may shrink the neighborhood so that $\widetilde{\kappa}_\nu \neq 0$ along the singular curve. By Proposition 5.6, it follows that

$$\beta \equiv 0 \quad \Longleftrightarrow \quad \omega_{3,4} \equiv 0$$

along the singular curve near p. Under this condition, Proposition 5.7 gives

$$\delta(p) \neq 0 \quad \Longleftrightarrow \quad \Pi_{3,5}(p) \neq 0 \quad \Longleftrightarrow \quad \widehat{\omega}_{3,5}(p) \neq 0 \quad \Longleftrightarrow \quad \omega_{3,5}(p) \neq 0.$$

The equivalence for $(3,5)$-cuspidal edges now follows from Theorem 5.12. □

Corollary 5.14. *Let $f : \Sigma \to \mathbb{R}^3$ and $g : \overline{\Sigma} \to \mathbb{R}^3$ be isometric frontals, and let $p \in \Sigma$ and $\overline{p} \in \overline{\Sigma}$ be corresponding points under a local isometry φ of their first fundamental forms, with $\varphi(p) = \overline{p}$. Assume that f is generic at p and g is generic at $\overline{p}$. Then, for $j = 4, 5$, the following statements are equivalent:*

(1) *f has a $(3,j)$-cuspidal edge at p.*

(2) *g has a $(3,j)$-cuspidal edge at $\overline{p}$.*

Proof. The first-kind condition and degeneracy order 2 are preserved under isometries. Moreover, β and δ are intrinsic. Thus, the result follows from Theorem 5.13. □

Example 5.15. Consider the map

$$f(u,v) = \left(u, \frac{u^2}{2} + \frac{v^3}{6}, \frac{v^4}{24}\right).$$

This map has a $(3,4)$-cuspidal edge at the origin, but $\widetilde{\kappa}_\nu = 0$ along the singular curve. Hence, $\Pi_{3,4} = \widetilde{\kappa}_\nu \omega_{3,4} = 0$, and consequently $\beta = 0$ since $\beta = 2\Pi_{3,4}$. Thus, in the non-generic case, β may vanish even at a $(3,4)$-cuspidal edge. This shows that the genericity assumption in Theorem 5.13 cannot be removed.

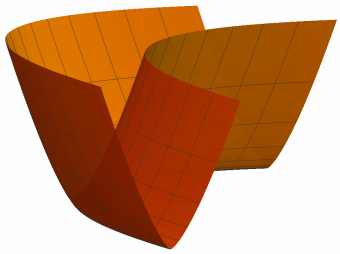


Figure 6: The image of $f(u,v) = \left(u, \frac{u^2}{2} + \frac{v^3}{6}, \frac{v^4}{24}\right)$. Although $\beta = 0$ since $\widetilde{\kappa}_\nu = 0$ along the singular curve, every singular point is a $(3,4)$-cuspidal edge.

Statements and Declarations

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Data availability

Data sharing not applicable to this article as no datasets were generated or analysed during the current study.

Conflict of interest

The authors declare that they have no conflict of interest.

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